

penev_B: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00001	1	1.000	$\left\langle \mathbf{s}_{\parallel}, \mathbf{s}_{\perp} \right\rangle = 0$	$\langle \mathbf{s}_{\parallel}, \mathbf{s}_{\perp} \rangle = 0$	from (4.57), $\langle \mathbf{s}_{\parallel}, \mathbf{s}_{\perp} \rangle = 0$, hence $\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$ (4.44). Analogously, the
penev_B_F00002	1	1.000	$\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$	$\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$	from (4.57), $\langle \mathbf{s}_{\parallel}, \mathbf{s}_{\perp} \rangle = 0$, hence $\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$ (4.44). Analogously, the
penev_B_F00003	1	1.000	O_m	O_m	Notably, the change of a single O_m does not change any of the other projections
penev_B_F00004	1	0.430	$\mathbf{s}^{\text{rec}}$	$\mathbf{s}^{\text{rec}}$	of $\mathbf{s}^{\text{rec}}$ to $\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$ (4.56); this comes directly from (4.54).
penev_B_F00005	1	0.430	$\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$	of $\mathbf{s}^{\text{rec}}$ to $\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$ (4.56); this comes directly from (4.54).
penev_B_F00006	1	0.577	$O^{\text{rec}}(\mathbf{x})$ (4.45)	$O^{\text{rec}}(\mathbf{x})$ (4.45)	Now we are prepared to calculate $O^{\text{rec}}(\mathbf{x})$ (4.45):
penev_B_F00007	1	0.606	$\{\mathbf{p}_m\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{\mathbf{p}_m\}_{\mathbf{x}_m \in \mathcal{M}}$	with the action of $\mathbf{O}^*$ on the dual-basis vectors $\{\mathbf{p}_m\}_{\mathbf{x}_m \in \mathcal{M}}$, only this time, the
penev_B_F00008	1	0.486	$\mathbf{Q}_{nm} = \langle \mathbf{q}_n, \mathbf{q}_m \rangle$	$\mathbf{Q}_{nm} = \langle \mathbf{q}_n, \mathbf{q}_m \rangle$	with $\mathbf{Q}_{nm} = \langle \mathbf{q}_n, \mathbf{q}_m \rangle$. Therefore, $\mathbf{B} = \mathbf{Q}^{-1}$ and
penev_B_F00009	1	0.486	$\mathbf{B} = \mathbf{Q}^{-1}$	$\mathbf{B} = \mathbf{Q}^{-1}$	with $\mathbf{Q}_{nm} = \langle \mathbf{q}_n, \mathbf{q}_m \rangle$. Therefore, $\mathbf{B} = \mathbf{Q}^{-1}$ and
penev_B_F00011	1	0.592	$\mathbf{Q}$	$\mathbf{Q}$	Notably, the matrix $\mathbf{Q}$ is the <i>restriction</i> of the matrix $\mathbf{P}$ on the set of reference
penev_B_F00012	1	0.592	$\mathbf{P}$	$\mathbf{P}$	Notably, the matrix $\mathbf{Q}$ is the <i>restriction</i> of the matrix $\mathbf{P}$ on the set of reference
penev_B_F00013	1	1.000	$\mathcal{M}$	$\mathcal{M}$	points, $\mathcal{M}$:
penev_B_F00014	1	0.826	$\mathbf{x}$	$\mathbf{x}$	Analogously to (4.48–4.49), the $\mathbf{x}$ dependence of the reconstructors $a_m(\mathbf{x})$ de-
penev_B_F00015	1	0.826	$a_m(\mathbf{x})$	$a_m(\mathbf{x})$	Analogously to (4.48–4.49), the $\mathbf{x}$ dependence of the reconstructors $a_m(\mathbf{x})$ de-
penev_B_F00016	1	1.000	ϕ	ϕ	depends only on the locations of the reference points $\mathcal{M}$, but not on ϕ itself.
penev_B_F00017	1	0.999	ϕ is ³⁵	ϕ is ³⁵	The reconstruction of the example ϕ is ³⁵
penev_B_F00018	1	0.654	$K^{(-1)}_n(\mathbf{x}) = \langle \hat{\mathbf{x}}, K^{(-1)} \hat{\mathbf{x}}_n \rangle$	$K^{(-1)}_n(\mathbf{x}) = \langle \hat{\mathbf{x}}, K^{(-1)} \hat{\mathbf{x}}_n \rangle$	with $K^{(-1)}_n(\mathbf{x}) = \langle \hat{\mathbf{x}}, K^{(-1)} \hat{\mathbf{x}}_n \rangle$.
penev_B_F00019	1	0.269	$ \mathcal{M} $	$ \mathcal{M} $	this might not seem natural—when $ \mathcal{M} $ is large, $\mathbf{Q}$ is also large, and inverting
penev_B_F00020	1	0.999	$\mathbf{Q}$	$\mathbf{Q}$	it is difficult. Nevertheless, $\mathbf{Q}$ is a special matrix—most of its values are van-
penev_B_F00021	1	0.623	$\{P_m(\mathbf{x})\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{P_m(\mathbf{x})\}_{\mathbf{x}_m \in \mathcal{M}}$	ishingly small; because the supports of $\{P_m(\mathbf{x})\}_{\mathbf{x}_m \in \mathcal{M}}$ are local, $\mathbf{Q}$ is <i>sparse</i> .
penev_B_F00022	1	0.990	$\mathbf{x}_m$	$\mathbf{x}_m$	For each point $\mathbf{x}_m$, the m -th column of $\mathbf{Q}$ has substantially nonvanishing val-
penev_B_F00023	1	0.990	m	m	For each point $\mathbf{x}_m$, the m -th column of $\mathbf{Q}$ has substantially nonvanishing val-
penev_B_F00024	1	0.446	$P_m(\mathbf{x}_{mn})$	$P_m(\mathbf{x}_{mn})$	$P_m(\mathbf{x}_m)$. Moreover, this is true even when the volume of the receptor grid V
penev_B_F00025	1	0.446	V	V	$P_m(\mathbf{x}_m)$. Moreover, this is true even when the volume of the receptor grid V
penev_B_F00026	1	0.961	$\mathbf{x}_m$	$\mathbf{x}_m$	increases—adding more and more receptors distant to $\mathbf{x}_m$ does not add non-
penev_B_F00027	1	0.905	N	N	small, regardless of N and V . ³⁶
penev_B_F00028	1	0.905	V is ³⁶	V is ³⁶	small, regardless of N and V . ³⁶
penev_B_F00029	1	0.709	$\phi(\mathbf{x})$	$\phi(\mathbf{x})$	In this Section, we have shown how to represent a sensory stimulus $\phi(\mathbf{x})$
penev_B_F00030	1	1.000	$\mathcal{P}[\phi(\mathbf{x})]$	$\mathcal{P}[\phi(\mathbf{x})]$	and its probability $\mathcal{P}[\phi(\mathbf{x})]$, in the context of an ensemble with low dimension-
penev_B_F00031	1	0.968	$O^{\text{rec}}(\mathbf{x})$	$O^{\text{rec}}(\mathbf{x})$	will result in a large error in $O^{\text{rec}}(\mathbf{x})$.
penev_B_F00032	1	0.743	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$	So far, we have addressed the question, given a sparsification $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$,
penev_B_F00033	1	1.000	$O(\mathbf{x})$	$O(\mathbf{x})$	how to reconstruct $O(\mathbf{x})$ and, from there, the example ϕ and its probabil-
penev_B_F00034	1	1.000	$\mathcal{P}[\phi]$	$\mathcal{P}[\phi]$	ity $\mathcal{P}[\phi]$.
penev_B_F00035	1	0.999	$\mathcal{M}^{(0)} = \emptyset$	$\mathcal{M}^{(0)} = \emptyset$	In the sparsification stage, we start with the empty set $\mathcal{M}^{(0)} = \emptyset$ and
penev_B_F00036	1	0.866	n	n	perform a series of iterative steps; at the n -th step, we add to $\mathcal{M}^{(n)}$ one reference
penev_B_F00037	1	0.866	$\mathcal{M}^{(n)}$	$\mathcal{M}^{(n)}$	perform a series of iterative steps; at the n -th step, we add to $\mathcal{M}^{(n)}$ one reference
penev_B_F00038	1	0.806	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$	The resulting set $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$ is an LFA representation of ϕ .
penev_B_F00039	1	1.000	$\mathcal{M} = \mathcal{M}^{(n)}$	$\mathcal{M} = \mathcal{M}^{(n)}$	First, given the current set $\mathcal{M} = \mathcal{M}^{(n)}$, we calculate the current maximum-
penev_B_F00040	1	0.479	$O^{\text{rec}}(\mathbf{x})$	$O^{\text{rec}}(\mathbf{x})$	likelihood estimate $O^{\text{rec}}(\mathbf{x})$ by (4.67–4.68) and calculate the current error

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penev_B_F00041	1	0.718	x_{m+1}	$\mathbf{x}_{m+1}$	Second, we look for the grid point $\mathbf{x}_{m+1}$, such that the value of $O^{\mathrm{err}}(\mathbf{x}_{m+1})$
penev_B_F00042	1	0.718	$0^{\{\text{err}\}}\left(\mathrm{x}_{m+1}\right)$	$O^{\mathrm{err}}(\mathbf{x}_{m+1})$	Second, we look for the grid point $\mathbf{x}_{m+1}$, such that the value of $O^{\mathrm{err}}(\mathbf{x}_{m+1})$
penev_B_F00043	1	0.980	$0^{\{\text{err}\}}\left(\mathbf{x}_{m+1}\right)^2$	$O^{\mathrm{err}}(\mathbf{x}_{m+1})^2$	is maximum. If its square, $O^{\mathrm{err}}(\mathbf{x}_{m+1})^2$, is below a predetermined threshold,
penev_B_F00044	1	1.000	$-\log \mathcal{P}[\phi]$	$-\log \mathcal{P}[\phi]$	terion. We will remember that the information content of the example, $-\log \mathcal{P}[\phi]$, is de-
penev_B_F00045	1	0.810	$-\log \mathcal{P}[\phi]-\left(-\log \mathcal{P}\left[\phi^{\mathrm{rec}}\right]\right)$	$-\log \mathcal{P}[\phi]-\left(-\log \mathcal{P}\left[\phi^{\mathrm{rec}}\right]\right)$	$-\log \mathcal{P}[\phi]-\left(-\log \mathcal{P}[\phi^{\mathrm{rec}}]\right)$, is equal to $\ O^{\mathrm{err}}\ _F^2$. Therefore, by bounding of the maximum
penev_B_F00046	1	0.810	$0^{\{\text{err}\}}\left _{U}^2\right.$	$\left\ O^{\mathrm{err}}\right\ _U^2$	$-\log \mathcal{P}[\phi]-\left(-\log \mathcal{P}[\phi^{\mathrm{rec}}]\right)$, is equal to $\ O^{\mathrm{err}}\ _F^2$. Therefore, by bounding of the maximum
penev_B_F00047	1	0.771	$0^{\{\text{err}\}}\left _E^2=\frac{1}{V} \sum_x O^{\mathrm{err}}(\mathbf{x})^2(2.8)\right.$	$\left\ O^{\mathrm{err}}\right\ _E^2=\frac{1}{V} \sum_x O^{\mathrm{err}}(\mathbf{x})^2(2.8)$	term in the sum $\left\ O^{\mathrm{err}}\right\ _E^2=\frac{1}{V} \sum_x O^{\mathrm{err}}(\mathbf{x})^2$ (2.8), we bound the whole sum and, therefore,
penev_B_F00048	1	0.779	$\mathbf{x}_{m+1}$	$\mathbf{x}_{m+1}$	Third, if the algorithm has not terminated yet, we add $\mathbf{x}_{m+1}$ to $\mathcal{M}^{(n)}$, which
penev_B_F00049	1	0.779	$\mathcal{M}^{\{(\mathbf{n})\}}$	$\mathcal{M}^{(\mathbf{n})}$	Third, if the algorithm has not terminated yet, we add $\mathbf{x}_{m+1}$ to $\mathcal{M}^{(n)}$, which
penev_B_F00050	1	0.974	$\mathcal{M}^{\{(n+1)\}}$	$\mathcal{M}^{(n+1)}$	results in $\mathcal{M}^{(n+1)}$, and go to step $n+1$. ³⁹
penev_B_F00051	1	0.974	$n+1 .^{\{39\}}$	$n+1 .^{\{39\}}$	results in $\mathcal{M}^{(n+1)}$, and go to step $n+1$. ³⁹
penev_B_F00052	1	0.978	$ \mathcal{M} =N$	$ \mathcal{M} =N$	dicted well, therefore they are linearly independent, and by the time $ \mathcal{M} =N$,
penev_B_F00053	1	0.972	$0^{\{\text{err}\}}(x)=0$	$O^{\mathrm{err}}(x)=0$	$O^{\mathrm{err}}(\mathbf{x})=\mathbf{0}$ (4.36).
penev_B_F00054	1	1.000	$\mathcal{M}^{\{(25)\}}$	$\mathcal{M}^{(25)}$	The location and the order of first 25 points, those in $\mathcal{M}^{(25)}$ (Fig. 4.4a),
penev_B_F00055	1	0.990	$\mathcal{M}^{\{(64)\}}$	$\mathcal{M}^{(64)}$	which is a desirable property. The points in $\mathcal{M}^{(64)}$ enough for acceptable
penev_B_F00056	1	0.822	a_r	a_r	number of values 64. We calculated the PCA coefficients a_r (2.47–2.52) and
penev_B_F00057	1	0.996	$\mathbf{x}_{m+1}$	$\mathbf{x}_{m+1}$	footnote 39 we argue that this is a costly calculation. Moreover, if we add the point $\mathbf{x}_{m+1}$,
penev_B_F00058	1	0.958	$0^{\{\text{err}\}}\left(\mathbf{x}_{m+1}\right)$	$O^{\mathrm{err}}(\mathbf{x}_{m+1})$	we are making a correction, $O^{\mathrm{err}}(\mathbf{x}_{m+1})$, to the estimated value $O^{\mathrm{rec}}(\mathbf{x}_{m+1})$; this has to be
penev_B_F00059	1	0.958	$0^{\{\text{rec}\}}\left(\mathbf{x}_{m+1}\right)$	$O^{\mathrm{rec}}(\mathbf{x}_{m+1})$	we are making a correction, $O^{\mathrm{err}}(\mathbf{x}_{m+1})$, to the estimated value $O^{\mathrm{rec}}(\mathbf{x}_{m+1})$; this has to be
penev_B_F00060	1	0.637	$O\left(\mathbf{x}_{m+1}\right)$	$O(\mathbf{x}_{m+1})$	compared to the error in the measurement of $O(\mathbf{x}_{m+1})$ itself, so a criterion defined at the
penev_B_F00061	1	0.938	$0^{\{\text{err}\}}\left(\mathbf{x}_{m+1}\right)$	$O^{\mathrm{err}}(\mathbf{x}_{m+1})$	level of the individual error, $O^{\mathrm{err}}(\mathbf{x}_{m+1})$, might not be such a bad idea after all. Also, we
penev_B_F00062	1	0.515	$0^{\{\text{err}\}}\left _U^2\right.$	$\left\ O^{\mathrm{err}}\right\ _U^2$	provisionally taken; $\mathbf{Q}$, inverted; the resulting $\left\ O^{\mathrm{err}}\right\ _U^2$, stored; and, after all $\sim V$ provisional
penev_B_F00063	1	0.515	$\sim V$	$\sim V$	provisionally taken; $\mathbf{Q}$, inverted; the resulting $\left\ O^{\mathrm{err}}\right\ _U^2$, stored; and, after all $\sim V$ provisional
penev_B_F00066	2	1.000	a	a	Ensemble 1. $N=220$, $n=\sigma_{400}$ (see Fig. 4.1). a : the first 25 points— $\mathcal{M}^{(25)}$,
penev_B_F00067	2	0.999	$\phi(\mathbf{x})$	$\phi(\mathbf{x})$	overlayed on $\phi(\mathbf{x})$ and numbered sequentially. (The label for point 11, in the
penev_B_F00068	2	0.999	b	b	is the bottommost point on the right edge, shown on b .) b : the points in $\mathcal{M}^{(64)}$.
penev_B_F00069	2	0.995	$N=220$	$N=220$	the LFA ones $O(\mathbf{x})$ (4.31) for $N=220$, then sparsified to get $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$.
penev_B_F00070	2	0.995	$\phi^{\mathrm{rec}}(\mathbf{x})$	$\phi^{\mathrm{rec}}(\mathbf{x})$	We calculated the reconstruction $\phi^{\mathrm{rec}}(\mathbf{x})$ either (a), from (2.50) with $N=64$, or
penev_B_F00071	2	0.995	$N=64$	$N=64$	We calculated the reconstruction $\phi^{\mathrm{rec}}(\mathbf{x})$ either (a), from (2.50) with $N=64$, or
penev_B_F00072	2	0.965	8×8	8×8	For comparison, we subsampled $\phi(\mathbf{x})$ on a uniform 8×8 grid and reconstructed
penev_B_F00073	2	1.000	Ψ_1	Ψ_1	it with those 64 points (c). In all cases we reconstructed about Ψ_1 the average
penev_B_F00074	2	1.000	a_1	a_1	of the ensemble; we kept a_1 fixed. Evidentially, the perceptual quality of the
penev_B_F00075	2	0.999	$\phi^{\mathrm{rec}}(\mathbf{x})$	$\phi^{\mathrm{rec}}(\mathbf{x})$	reconstruction $\phi^{\mathrm{rec}}(\mathbf{x})$ (top row) is the best, and the error $\phi(\mathbf{x})-\phi^{\mathrm{rec}}(\mathbf{x})$ (bot-
penev_B_F00076	2	0.999	$\phi(\mathbf{x})-\phi^{\mathrm{rec}}(\mathbf{x})$	$\phi(\mathbf{x})-\phi^{\mathrm{rec}}(\mathbf{x})$	reconstruction $\phi^{\mathrm{rec}}(\mathbf{x})$ (top row) is the best, and the error $\phi(\mathbf{x})-\phi^{\mathrm{rec}}(\mathbf{x})$ (bot-
penev_B_F00077	2	0.499	c	c	and 508 for (a), (b), and (c), respectively, out of 560 total power in $(\phi(\mathbf{x})-a_1 \Psi_1)$. Notably,
penev_B_F00078	2	0.499	$(\phi(\mathbf{x})-a_1 \Psi_1)$	$(\phi(\mathbf{x})-a_1 \Psi_1)$	and 508 for (a), (b), and (c), respectively, out of 560 total power in $(\phi(\mathbf{x})-a_1 \Psi_1)$. Notably,
penev_B_F00079	2	0.990	Ψ_2	Ψ_2	as Ψ_2 on Fig. 3.9. Indeed, the m.s.e. due to the error in a_2 in (b) is 75. If one fixes not
penev_B_F00080	2	0.990	a_2	a_2	as Ψ_2 on Fig. 3.9. Indeed, the m.s.e. due to the error in a_2 in (b) is 75. If one fixes not
penev_B_F00088	2	1.000	$(m-1)$	$(m-1)$	the $(m-1)$ -st step through a simple algebraic formula. The algorithm for inversion through
penev_B_F00092	2	0.560	99%	99%	to get 99% of the fidelity?—the incremental benefit of adding a reference point in the 50%
penev_B_F00093	2	0.560	50%	50%	to get 99% of the fidelity?—the incremental benefit of adding a reference point in the 50%
penev_B_F00094	2	1.000	r	r	LFA representations by taking the sums in (4.29) in several separate bands for r , instead of
penev_B_F00101	3	1.000	$\mathcal{M}^{\{(50)\}}$	$\mathcal{M}^{(50)}$	on Fig. 4.8. Almost all of the points in the respective sets $\mathcal{M}^{(50)}$ are on the
penev_B_F00102	3	0.496	$0\left(x_m\right)_{x_m \in \mathcal{M}}$	$\{O\left(x_m\right)\}_{x_m \in \mathcal{M}}$	given a sparsification $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$ of the dense LFA outputs $O(\mathbf{x})$ for an
penev_B_F00103	3	0.496	$O(x)$	$O(x)$	given a sparsification $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$ of the dense LFA outputs $O(\mathbf{x})$ for an
penev_B_F00104	3	0.951	$\left\{a_m(\mathbf{x})\right\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{a_m(\mathbf{x})\}_{\mathbf{x}_m \in \mathcal{M}}$	given by the set of interpolating reconstructors $\{a_m(\mathbf{x})\}_{\mathbf{x}_m \in \mathcal{M}}$ (4.68).
penev_B_F00105	3	0.342	$\sum_{n=1}^{\mathcal{M}} \mathbf{Q}^{-1} n n n K^{(-1)} n(\mathbf{x})$	$\sum_{n=1}^{\mathcal{M}} \mathbf{Q}^{-1} n n n K^{(-1)} n(\mathbf{x})$	combination, only with a different set of reconstructors, $\sum_{n=1}^{\mathcal{M}} \mathbf{Q}^{-1} n n n K^{(-1)} n(\mathbf{x})$.
penev_B_F00108	3	0.938	$\mathbf{Q}^{-1}$	$\mathbf{Q}^{-1}$	operation. By the same argument, $\mathbf{Q}^{-1}$ will be sparse, and acting with it on O_m
penev_B_F00109	3	0.907	$\mathcal{O}_I$	$\mathcal{O}_I$	will be a local operation too. Moreover, one may try to solve for $\mathcal{O}_I$ directly
penev_B_F00110	3	1.000	$\mathbf{Q}^{-1}$	$\mathbf{Q}^{-1}$	and never explicitly store $\mathbf{Q}^{-1}$, which will eliminate one computational step
penev_B_F00111	3	0.644	$\mathcal{O}_1$	$\mathcal{O}_1$	Some intuition into what is involved in the calculation of $\mathcal{O}_1$ may be de-
penev_B_F00112	3	1.000	$\mathcal{O}_l$	$\mathcal{O}_l$	may be solved for $\mathcal{O}_l$ (cf. 4.72) by a rapid iteration.

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penev_B_F00113	4	0.601	$G(\mathrm{cal}\{0\})(\mathrm{bf}\{x\})=\mathrm{mathrm}\{g\}(\mathrm{cal}\{0\}(\mathrm{bf}\{x\}))$	$G(\mathcal{O})(\mathbf{x}) = g(\mathcal{O}(\mathbf{x}))$	where $\boxed{G(\mathcal{O})(\mathbf{x}) = g(\mathcal{O}(\mathbf{x}))}$ with g , a suitably chosen <i>sigmoid</i> function, ⁴⁵ and
penev_B_F00114	4	1.000	τ	τ	τ is the characteristic time scale.
penev_B_F00115	4	1.000	$\mathrm{cal}\{0\}(\mathrm{bf}\{x\})$	$\mathcal{O}(\mathbf{x})$	to pass the large ones intact. Then, if all $\mathcal{O}(\mathbf{x})$ either vanish, or are sufficiently
penev_B_F00116	4	0.555	$\frac{d}{d\,t}\,\mathrm{cal}\{0\}(\mathrm{bf}\{x\})=0$	$\frac{d}{dt}\mathcal{O}(\mathbf{x}) = 0$	large, g will be the identity. In this regime, for the steady states, $\frac{d}{dt}\mathcal{O}(\mathbf{x}) = 0$,
penev_B_F00117	4	0.465	$0^{\{\mathrm{rm}\{\mathrm{rec}\}}=\mathrm{mathrm}\{P\}\,\mathrm{cal}\{0\}=\mathrm{mathrm}\{K\}\,\phi=0$	$O^{\mathrm{rec}} = P\mathcal{O} = K\phi = O$	The steady-state equation (4.76) is equivalent to $O^{\mathrm{rec}} = P\mathcal{O} = K\phi = O$
penev_B_F00118	4	0.997	$\left\{\mathrm{cal}\{0\}_{\{m\}}\right\}_{x_m\in\mathrm{cal}\{M\}}$	$\{\mathcal{O}_m\}_{x_m\in\mathcal{M}}$	hence, the sparsified outputs $\{\mathcal{O}_m\}_{\mathbf{x}_m\in\mathcal{M}}$ are solutions of (4.76) and, therefore,
penev_B_F00119	4	0.840	$\mathrm{mathrm}\{g\}:\mathrm{cal}\{R\}\rightarrow[a,\,b]$	$g:\mathcal{R}\rightarrow[a,b]$	⁴⁶ A sigmoid function is a monotonic nondecreasing function $g:\mathcal{R}\rightarrow[a,b]$, often $a=-b$,
penev_B_F00120	4	0.840	$a=-b$	$a=-b$	⁴⁶ A sigmoid function is a monotonic nondecreasing function $g:\mathcal{R}\rightarrow[a,b]$, often $a=-b$,
penev_B_F00121	4	0.809	$\mathrm{mathrm}\{g\}^{\{\prime\}}$	g'	and usually g' is non-zero only near the origin, or a suitably chosen <i>threshold</i> θ .
penev_B_F00122	4	0.809	θ	θ	and usually g' is non-zero only near the origin, or a suitably chosen <i>threshold</i> θ .
penev_B_F00131	4	0.800	g	g	If output dynamic range equalization is required, g may be chosen to preclude
penev_B_F00132	4	0.607	$\mathrm{cal}\{0\}_{\{i\}}$	$\mathcal{O}_i$	large $\mathcal{O}_i$.
penev_B_F00133	4	0.863	$\mathrm{cal}\{P\}_{\{\text{objects}\}}$	$\mathcal{P}_{\mathrm{objects}}$	ability distribution, $\mathcal{P}_{\mathrm{objects}}$, is entirely contained in the probability distribution
penev_B_F00134	4	0.992	$\mathrm{cal}\{P\}_{\{\text{images}\}}$	$\mathcal{P}_{\mathrm{images}}$	of natural images, $\mathcal{P}_{\mathrm{images}}$.
penev_B_F00135	4	0.445	$\mathrm{cal}\{P\}_{\{\text{object}\}}\,\mathrm{mathrm}\{\mathrm{can}\}$	$\mathcal{P}_{\mathrm{object\,can}}$	approximation and parameterization. In that sense, $\mathcal{P}_{\mathrm{objects}}$ can be considered
penev_B_F00136	5	0.945	$\left\{I^t\right\}_{t\in T}$	$\{I^t\}_{t\in T}$	of images $\{I^t\}_{t\in T}$ was prepared, wherein each image contained a human face
penev_B_F00137	5	1.000	$I^{\{t\}}$	I^t	in a relatively frontal pose. For each image in the ensemble, I^t , the locations
penev_B_F00138	5	1.000	$\left(X_{\{l\}}^{\{t\}},\,Y_{\{l\}}^{\{t\}}\right)$	(X_l^t, Y_l^t)	(X_l^t, Y_l^t) and (X_r^t, Y_r^t) of the left and right eye, respectively (see discussion in
penev_B_F00139	5	1.000	$\left(X_{\{r\}}^{\{t\}},\,Y_{\{r\}}^{\{t\}}\right)$	(X_r^t, Y_r^t)	(X_l^t, Y_l^t) and (X_r^t, Y_r^t) of the left and right eye, respectively (see discussion in
penev_B_F00140	5	0.698	$\theta^{\{i\}}$	θ^i	Subsequently, the image I^t was <i>rotated</i> , by the angle θ^t , to line the eyes
penev_B_F00141	5	1.000	$S^{\{i\}}$	S^i	horizontally; <i>scaled</i> , by the scale factor S^t , to bring the inter-eye distance to a
penev_B_F00142	5	0.998	$X^{\{t\}}$	X^t	pre-defined length; and <i>translated</i> , horizontally by X^t and vertically by Y^t , to
penev_B_F00143	5	0.998	$Y^{\{t\}}$	Y^t	pre-defined length; and <i>translated</i> , horizontally by X^t and vertically by Y^t , to
penev_B_F00144	5	0.998	$\phi^{\{t\}}$	ϕ^t	for a given cropping, the example ϕ^t depends on the image I^t , as well as on the
penev_B_F00145	5	0.827	H	H	where H is the (highly nonlinear) operator of symmetry-breaking.
penev_B_F00146	5	0.554	$\left(X_{\{l\}},\,Y_{\{l\}}\right)$	(X_l, Y_l)	eye locations, (X_l, Y_l) and $\boxed{(X_r, Y_r)}$, lead to errors in α . Also, if $H(I^t; \alpha^t)$ is in
penev_B_F00147	5	0.554	$\left(X_{\{r\}},\,Y_{\{r\}}\right)$	(X_r, Y_r)	eye locations, (X_l, Y_l) and (X_r, Y_r) , lead to errors in α . Also, if $H(I^t; \alpha^t)$ is in
penev_B_F00148	5	0.554	α	α	eye locations, (X_l, Y_l) and (X_r, Y_r) , lead to errors in α . Also, if $H(I^t; \alpha^t)$ is in
penev_B_F00149	5	0.554	$H\left(I^{\{t\}}\,;\,\alpha^{\{t\}}\right)$	$H\left(I^t; \alpha^t\right)$	eye locations, (X_l, Y_l) and (X_r, Y_r) , lead to errors in α . Also, if $H(I^t; \alpha^t)$ is in
penev_B_F00150	5	1.000	$H\left(I^{\{t\}}\,;\,\alpha^{\{t\}}+\Delta\alpha\right)$	$H\left(I^t; \alpha^t + \Delta\alpha\right)$	the ensemble of faces, then $H(I^t; \alpha^t + \Delta\alpha)$ will also be in it, provided that $\Delta\alpha$
penev_B_F00151	5	1.000	$\Delta\alpha$	$\Delta\alpha$	the ensemble of faces, then $H(I^t; \alpha^t + \Delta\alpha)$ will also be in it, provided that $\Delta\alpha$
penev_B_F00153	5	1.000	$\left\{\Psi_{\{r\}}(\mathrm{bf}\{x\})\right\}_{r=1}^{16}$	$\{\Psi_r(\mathbf{x})\}_{r=1}^{16}$	The first 16 eigenmodes $\{\Psi_r(\mathbf{x})\}_{r=1}^{16}$ of Ensemble 6 (top and middle) and eight
penev_B_F00154	5	0.998	2 A	$2A$	This figure is reproduced on Plate 2A. For comparison, part of Fig. 3.9 is
penev_B_F00155	5	1.000	2 B	$2B$	reproduced on Plate 2B.
penev_B_F00156	5	1.000	$\Psi_{\{8\}}$	Ψ_8	Notably, the first eyebrow mode is now very strong Ψ_8 . This is due to
penev_B_F00157	5	1.000	$\Psi_{\{12\}}$	Ψ_{12}	A very interesting effect on the asymmetric Ψ_{12} (Fig. 3.9), the first strong
penev_B_F00158	5	1.000	$\Psi_{\{13\}}$	Ψ_{13}	feature therein, can be observed—it has moved one mode down to Ψ_{13} in the
penev_B_F00159	5	1.000	$\Psi_{\{14\}}$	Ψ_{14}	symmetrized ensemble, has weakened, and has spread to two more modes: Ψ_{14}
penev_B_F00160	5	1.000	$\Psi_{\{18\}}$	Ψ_{18}	and Ψ_{18} . The same has happened to Ψ_{14} , the “Nixon” mode (footnote 11 on
penev_B_F00161	5	1.000	$\Psi_{\{14\}},\,\Psi_{\{15\}},\,\Psi_{\{16\}}$	$\Psi_{14}, \Psi_{15}, \Psi_{16}$	page 34) weak vestiges of it can be observed on Ψ_{14} , Ψ_{15} , Ψ_{16} , and Ψ_{18} .
penev_B_F00162	5	0.439	$\boldsymbol{\Psi}_{\{1\}}$	$\boldsymbol{\Psi}_1$	they are thin—it is evident from Ψ_1 and Ψ_{13} that the distance from the eyebrows to the eyes
penev_B_F00163	5	0.439	$\boldsymbol{\Psi}_{\{13\}}$	$\boldsymbol{\Psi}_{13}$	they are thin—it is evident from Ψ_1 and Ψ_{13} that the distance from the eyebrows to the eyes
penev_B_F00164	5	0.833	$\boldsymbol{\Psi}_{\{8\}}$	$\boldsymbol{\Psi}_8$	eyebrow location. This degree of freedom is what is observed on Ψ_8 .
penev_B_F00165	5	1.000	$\Psi_{\{100\}}$	Ψ_{100}	The feature localization problem is very clearly evident in Ψ_{100} and Ψ_{200} .
penev_B_F00166	5	1.000	$\Psi_{\{200\}}$	Ψ_{200}	The feature localization problem is very clearly evident in Ψ_{100} and Ψ_{200} .
penev_B_F00167	5	0.935	$\Psi_{\{500\}}$	Ψ_{500}	Interestingly, Ψ_{500} and Ψ_{1000} show quite discernible face content: even Ψ_{2006} .
penev_B_F00168	5	0.935	$\Psi_{\{1000\}}$	Ψ_{1000}	Interestingly, Ψ_{500} and Ψ_{1000} show quite discernible face content: even Ψ_{2006} .
penev_B_F00169	5	0.935	$\Psi_{\{2000\}}$	Ψ_{2000}	Interestingly, Ψ_{500} and Ψ_{1000} show quite discernible face content: even Ψ_{2006} .
penev_B_F00170	5	1.000	$N>60$	$N > 60$	is better for any $N > 60$. Also, the respective perceptual identity information
penev_B_F00171	5	0.955	$\phi_{1000}^{\mathrm{err}}\{\mathrm{e}\,\mathrm{r}\,\mathrm{r}\}$	$\phi_{1000}^{\mathrm{err}}$	in the error is smaller, almost nonexistent in $\phi_{1000}^{\mathrm{err}}$.
penev_B_F00172	5	0.987	$S-U$	$S-U$	S - $\mathcal{U}$ diagrams on Fig. 5.3. Not only does the symmetrized ensemble have
penev_B_F00173	5	0.994	$N\,\mathrm{approx}\,300$	$N \approx 300$	Whereas the entropic cost is rapidly growing after $N \approx 300$ for the asym-
penev_B_F00174	5	1.000	$N\,\mathrm{in}[200,500]$	$N \in [200, 500]$	metric ensemble, it is relatively flat in the regime $\boxed{N \in [200, 500]}$ for the sym-
penev_B_F00175	6	1.000	$\phi_N^{\mathrm{rec}}\{\text{rec}\}$	ϕ_N^{rec}	The successive reconstructions ϕ_N^{rec} labeled with their SNR (top) and their
penev_B_F00176	6	0.999	$\phi_N^{\mathrm{err}}\{\text{err}\}$	ϕ_N^{err}	respective residual errors ϕ_N^{err} (bottom) for Example 1 (Fig. 3.10- <i>original</i>).
penev_B_F00177	6	0.923	$\{\}^{\{6\}}\,\mathrm{mathrm}\{\mathrm{cf}\}\,.\,\Psi_{\{8\}}$	⁶ cf. Ψ_8	⁶ cf. Ψ_8 on Fig. 5.1

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00192	6	■ 0.974	H(I ; \alpha)(5.2)	$H(I; \alpha)(5.2)$	$H(I; \alpha)$ (5.2) on the set of images $\{I^t\}_{t \in T}$. In the restricted sense, used for the
penev_B_F00193	6	■ 0.974	\left\{\alpha^t\right\}_{t \in T}(5.1),\{ \}^{\wedge 8}	$\{\alpha^t\}_{t \in T}(5.1),^8$	of coefficients $\{\alpha^t\}_{t \in T}(5.1),^8$ and one example ϕ^t was included in the respective
penev_B_F00194	6	■ 1.000	I^{\wedge \{i\}}	I^i	ensembles for each image I^t .
penev_B_F00195	6	■ 0.336	\left\{\phi^{\wedge \{t\}}\left(\alpha^{\wedge \{t\}}\right)\right\}_{t \in T}	$\{\phi^t(\alpha^t)\}_{t \in T}$	the objects $\{\phi^t(\alpha^t)\}_{t \in T}$, but also all objects $\{\phi^t(\alpha)\}_{t \in T, \alpha \in \Delta_{\alpha^t}}$, where Δ_{α^t} is
penev_B_F00196	6	■ 0.336	\left\{\phi^{\wedge \{i\}}(\alpha)\right\}_{t \in T^{\wedge \prime}, \alpha \in \Delta_{\alpha^{\wedge i}}}	$\{\phi^i(\alpha)\}_{t \in T', \alpha \in \Delta_{\alpha^i}}$	the objects $\{\phi^t(\alpha^t)\}_{t \in T}$, but also all objects $\{\phi^t(\alpha)\}_{t \in T, \alpha \in \Delta_{\alpha^t}}$, where Δ_{α^t} is
penev_B_F00197	6	■ 0.336	\Delta_{\alpha^{\wedge i}}	Δ_{α^i}	the objects $\{\phi^t(\alpha^t)\}_{t \in T}$, but also all objects $\{\phi^t(\alpha)\}_{t \in T, \alpha \in \Delta_{\alpha^t}}$, where Δ_{α^t} is
penev_B_F00198	6	■ 0.986	\alpha^{\wedge \{t\}}	α^t	suitably chosen closely around α^t . In all cases, the probabilities $\mathcal{P}[\phi^t(\alpha)]$ were
penev_B_F00199	6	■ 0.986	\mathcal{P}\left[\phi^{\wedge \{t\}}(\alpha)\right]	$\mathcal{P}[\phi^t(\alpha)]$	suitably chosen closely around α^t . In all cases, the probabilities $\mathcal{P}[\phi^t(\alpha)]$ were
penev_B_F00200	6	■ 0.994	S	S	determined in the context of PCA by the projector on the PCA S -subspace $\mathbf{P}'$
penev_B_F00201	6	■ 0.994	\mathbf{P}^{\wedge \prime}	$\mathbf{P}'$	determined in the context of PCA by the projector on the PCA S -subspace $\mathbf{P}'$
penev_B_F00202	6	■ 0.966	I	I	given image I and an initial guess α_0 , calculate the entropic costs (2.49)
penev_B_F00203	6	■ 0.966	\alpha_{\{0\}}	α_0	given image I and an initial guess α_0 , calculate the entropic costs (2.49)
penev_B_F00204	6	■ 0.860	\left\{\phi(\alpha)\right\}_{\alpha \in \Delta_{\alpha_0}}	$\{\phi(\alpha)\}_{\alpha \in \Delta_{\alpha_0}}$	for all images $\{\phi(\alpha)\}_{\alpha \in \Delta_{\alpha_0}}$. The maximum likelihood estimate is α that mini-
penev_B_F00205	6	■ 0.769	\langle S\rangle(5.4)	$\langle S\rangle(5.4)$	mizes $\langle S\rangle$ (5.4) and, therefore, maximizes the probability $\mathcal{P}[\phi(\alpha)]$ (2.20).
penev_B_F00206	6	■ 0.769	\mathcal{P}[\phi(\alpha)]	$\mathcal{P}[\phi(\alpha)]$	mizes $\langle S\rangle$ (5.4) and, therefore, maximizes the probability $\mathcal{P}[\phi(\alpha)]$ (2.20).
penev_B_F00207	6	■ 0.998	\alpha^{\wedge 2}	α^2	the parameters of the symmetry breaking α^t . For the ensembles used in Chapter 3, the
penev_B_F00208	6	■ 0.723	\left\{\left(X_l^{\wedge t}, Y_l^{\wedge t} ; X_r^{\wedge t}, Y_r^{\wedge t}\right)\right\}_{t \in T}	$\{(X_l^t, Y_l^t; X_r^t, Y_r^t)\}_{t \in T}$	$\{(X_l^t, Y_l^t; X_r^t, Y_r^t)\}_{t \in T}$. From them, the wonderful set of high-quality coefficients $\{\alpha_{G, M}^t\}_{t \in T}$
penev_B_F00209	6	■ 0.723	\left\{\alpha_{G, M}^{\wedge t}\right\}_{t \in T}	$\{\alpha_{G, M}^t\}_{t \in T}$	$\{(X_l^t, Y_l^t; X_r^t, Y_r^t)\}_{t \in T}$. From them, the wonderful set of high-quality coefficients $\{\alpha_{G, M}^t\}_{t \in T}$
penev_B_F00210	7	■ 0.861	\langle S\rangle\left(N, \alpha_h^{\wedge t}\right)	$\langle S\rangle\left(N, \alpha_h^t\right)$	The average entropic cost $\langle S\rangle\left(N, \alpha_h^t\right)$ (5.4) is plotted as a function of N for 7
penev_B_F00211	7	■ 0.998	\alpha_h^{\wedge t}	α_h^t	different choices of α_h^t (5.5) for one of the images I^t in Ensemble 6 (see Ap-
penev_B_F00212	7	■ 0.999	\phi^{\wedge \{t\}}(0)(=)	$\phi^t(0)(=)$	The vertical axis is additionally scaled so that the in-sample example $\phi^t(0)(=)$
penev_B_F00213	7	■ 0.985	\langle S\rangle(N) \rightarrow \frac{\sigma_0^2}{\sum_{r=1}^N \sigma_r^2}	$\langle S\rangle(N) \rightarrow \frac{\sigma_0^2}{\sum_{r=1}^N \sigma_r^2}$	has $\langle S\rangle(N) \rightarrow \frac{\sigma_0^2}{\sum_{r=1}^N \sigma_r^2}$ when $N \rightarrow T$, instead of the more usual, 1.
penev_B_F00214	7	■ 0.985	N \rightarrow T	$N \rightarrow T$	has $\langle S\rangle(N) \rightarrow \frac{\sigma_0^2}{\sum_{r=1}^N \sigma_r^2}$ when $N \rightarrow T$, instead of the more usual, 1.
penev_B_F00215	7	■ 0.999	\phi^{\wedge t}\left(h_n\right)	$\phi^t\left(h_n\right)$	The six out-of-sample examples $\phi^t\left(h_n\right)$ were produced with $h_n=1-s_n, s_n=$
penev_B_F00216	7	■ 0.999	h_{\{n\}}=1-s_{\{n\}}, s_{\{n\}}=	$h_n=1-s_n, s_n=$	The six out-of-sample examples $\phi^t\left(h_n\right)$ were produced with $h_n=1-s_n, s_n=$
penev_B_F00217	7	■ 1.000	1.05^{\wedge \{n / 2\}}	$1.05^{n / 2}$	$1.05^{n / 2}$, for $n \in\{-3,-2,-1,1,2,3\}$, as outlined in Appendix A.1.
penev_B_F00218	7	■ 1.000	n \in\{-3,-2,-1,1,2,3\}	$n \in\{-3,-2,-1,1,2,3\}$	$1.05^{n / 2}$, for $n \in\{-3,-2,-1,1,2,3\}$, as outlined in Appendix A.1.
penev_B_F00219	7	■ 0.998	I^{\wedge \{t\}}, \alpha^{\wedge \{t\}}	I^t, α^t	for definition) are shown on Fig. 5.5. For all 1216 images I^t , α^t was determined.
penev_B_F00220	7	■ 1.000	\phi^{\wedge t}(h)=H\left(I^{\wedge t} ; \alpha_h^{\wedge t}\right)	$\phi^t(h)=H\left(I^t ; \alpha_h^t\right)$	From each α^t , several examples $\phi^t(h)=H\left(I^t ; \alpha_h^t\right)$ were produced with
penev_B_F00221	7	■ 1.000	\phi^{\wedge \{t\}}(0)\right\}	$\{\phi^t(0)\}$	as outlined in Appendix A.1. Only the examples $\phi^t(0)$ were used to derive
penev_B_F00222	7	■ 0.997	T=1216	$T=1216$	the modes, hence $T=1216$.
penev_B_F00223	7	■ 1.000	\langle S\rangle(N) \rightarrow 1	$\langle S\rangle(N) \rightarrow 1$	Two effects can be observed. First, the in-sample effect causes $\langle S\rangle(N) \rightarrow 1$
penev_B_F00224	7	■ 1.000	h=0	$h=0$	with $N \rightarrow T$ for $h=0$. Also, for smaller deviations from in-sample-ness $ h $,
penev_B_F00225	7	■ 1.000	h	$ h $	with $N \rightarrow T$ for $h=0$. Also, for smaller deviations from in-sample-ness $ h $,
penev_B_F00226	7	■ 0.509	\langle S\rangle\left(T_i \alpha_h\right)	$\langle S\rangle\left(T_i \alpha_h\right)$	the entropic cost of the un-truncated representation, $\langle S\rangle\left(T, \alpha_h\right)$, is generally
penev_B_F00227	7	■ 1.000	N \in[350,750]	$N \in[350,750]$	Second, for the “reasonable regime” $N \in[350,750]$, the out-of-sample ex-
penev_B_F00228	7	■ 0.999	\phi^{\wedge t}\left(h_{+1}\right)	$\phi^t\left(h_{+1}\right)$	ample $\phi^t\left(h_{+1}\right)$ is more probable than the in-sample example $\phi^t(0)$. Therefore,
penev_B_F00229	7	■ 0.999	\phi^{\wedge \{t\}}(0)	$\phi^t(0)$	ample $\phi^t\left(h_{+1}\right)$ is more probable than the in-sample example $\phi^t(0)$. Therefore,
penev_B_F00231	7	■ 1.000	n \in\{-4,-3,3,4\}	$n \in\{-4,-3,3,4\}$	the examples $\phi^t\left(h_n\right)$, with $n \in\{-4,-3,3,4\}$, have been substituted with the
penev_B_F00232	7	■ 0.996	\phi^{\wedge t}\left(h_m\right)	$\phi^t\left(h_m\right)$	examples $\phi^t\left(h_m\right)$ from Ensemble 7 (see Appendix A.1 for definition) with $h_m=$
penev_B_F00233	7	■ 0.996	h_{\{m\}}=	$h_m=$	examples $\phi^t\left(h_m\right)$ from Ensemble 7 (see Appendix A.1 for definition) with $h_m=$
penev_B_F00234	7	■ 0.990	1-s_{\{m\}}, s_{\{m\}}=1.025^{\wedge \{m / 3\}}	$1-s_m, s_m=1.025^{m / 3}$	$1-s_m, s_m=1.025^{m / 3}$. In this notation, $n=1$ produces almost the same
penev_B_F00235	7	■ 0.990	n=1	$n=1$	$1-s_m, s_m=1.025^{m / 3}$. In this notation, $n=1$ produces almost the same
penev_B_F00236	7	■ 1.000	m=3	$m=3$	example as $m=3$.
penev_B_F00237	7	■ 0.911	(--);-1(-=);+1(+=)	$(--);-1(-=);+1(+=)$	The values of m are: -2 , labeled with $(--)$; -1 $(-=)$; $+1$ $(+=)$; and $+2$ $(++)$.
penev_B_F00238	7	■ 0.911	+2(++)	$+2(++)$	The values of m are: -2 , labeled with $(--)$; -1 $(-=)$; $+1$ $(+=)$; and $+2$ $(++)$.
penev_B_F00239	7	■ 0.813	\alpha_{+1}^{\wedge t}	α_{+1}^t	for a reasonable truncation, α_{+1}^t is a better parameter of symmetry breaking
penev_B_F00240	7	■ 1.000	\alpha^{\wedge \{i\}}	α^i	than α^t .
penev_B_F00241	7	■ 1.000	h	h	optimal value for h .
penev_B_F00243	7	■ 0.963	N, \langle S\rangle\left(N, \alpha_h^{\wedge t}\right)	$N, \langle S\rangle\left(N, \alpha_h^t\right)$	Evidently, for all N , $\langle S\rangle\left(N, \alpha_h^t\right)$ has a local minimum for $h>0$. As N
penev_B_F00244	7	■ 0.963	h>0	$h>0$	Evidently, for all N , $\langle S\rangle\left(N, \alpha_h^t\right)$ has a local minimum for $h>0$. As N
penev_B_F00245	7	■ 1.000	h \approx+1.5 \%	$h \approx+1.5 \%$	increases, this minimum is achieved for $h \approx+1.5 \%$, and is forced to $h=0$ as
penev_B_F00246	7	■ 0.998	h=1.5 \%	$h=1.5 \%$	the in-sample effect takes over. Therefore, one can conclude that $h=1.5 \%$
penev_B_F00247	7	■ 1.000	T>N	$T>N$	which will result in $T>N$), there will be one optimal value of h for all N . A

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00248	7	0.777	$\phi^i(h)$	$\phi^i(h)$	studied $\phi^t(h)$. In this manner, for each image I^t , we produced the <i>trajectory</i>
penev_B_F00249	7	1.000	$\left\{ \Phi^{*+} \phi^t(h) \right\} \subset S$	$\{ \Phi^{*+} \phi^t(h) \} \subset S$	$\{ \Phi^{*+} \phi^t(h) \} \subset S$.
penev_B_F00250	7	0.998	$\left\{ \Phi^{*+} \phi^t(h) \right\} \subset S^2$	$\ \Phi^{*+} \phi^t(h) \ _S^2$	jectory and the origin, $\ \Phi^{*+} \phi^t(h) \ _S^2$ as the entropy (2.20) of the example $\phi^t(h)$
penev_B_F00251	7	0.998	$\phi^t(h)$	$\phi^t(h)$	jectory and the origin, $\ \Phi^{*+} \phi^t(h) \ _S^2$ as the entropy (2.20) of the example $\phi^t(h)$
penev_B_F00252	7	1.000	k	k	In general, α has k , more than one, independent components for the
penev_B_F00253	7	0.937	$\phi^l(\alpha)$	$\phi^l(\alpha)$	special case discussed in Section 5.1 they are four (5.1)—and $\phi^t(\alpha)$ is a k -
penev_B_F00254	7	0.941	$\Phi^{*+} \phi(\alpha)$	$\Phi^{*+} \phi(\alpha)$	points on the manifold $\Phi^{*+} \phi(\alpha)$ is a function of $\alpha \rightarrow S(\alpha)$. On one hand, for
penev_B_F00255	7	0.941	$\alpha - S(\alpha)$	$\alpha - S(\alpha)$	points on the manifold $\Phi^{*+} \phi(\alpha)$ is a function of $\alpha \rightarrow S(\alpha)$. On one hand, for
penev_B_F00256	7	1.000	$\phi(\alpha)$	$\phi(\alpha)$	values of α that are very wrong, the probability that $\phi(\alpha)$ is in the ensemble
penev_B_F00257	7	0.996	$S(\alpha)$	$S(\alpha)$	will be very low and $S(\alpha)$ will be very large a tree is not a face, as well as
penev_B_F00258	7	0.995	S_{max}	S_{max}	one can find a reasonable threshold S_{max} and a finite compact region on the
penev_B_F00259	7	0.904	Δ	Δ	manifold Δ , such that $\{ \alpha S(\alpha) < S_{\text{max}} \} \subset \Delta$.
penev_B_F00260	7	0.904	$\left\{ \alpha \mid S(\alpha) < S_{\text{max}} \right\} \subset \Delta$	$\{ \alpha \mid S(\alpha) < S_{\text{max}} \} \subset \Delta$	manifold Δ , such that $\{ \alpha S(\alpha) < S_{\text{max}} \} \subset \Delta$.
penev_B_F00261	8	0.787	α_{GM}^t	α_{GM}^t	and for a reasonable choice for N , and for a reasonable initial estimate α_{GM}^t .
penev_B_F00262	8	1.000	$N \approx 10$	$N \approx 10$	Suppose that the knowledge extends only up to $N \approx 10$. In that context, only
penev_B_F00263	8	0.791	X, Y	X, Y	contact, the example is centered at the fovea and, therefore, the (X, Y) part
penev_B_F00264	8	1.000	Ψ_3	Ψ_3	due to the strong “Lambertian” mode Ψ_3 (Fig. 3.9) and its symmetrizations.
penev_B_F00265	8	0.893	Ψ_2, Ψ_4	Ψ_2, Ψ_4	that other strong irrelevant modes—the background modes Ψ_2, Ψ_4 , and Ψ_6 ,
penev_B_F00266	8	0.893	Ψ_5	Ψ_5	that other strong irrelevant modes—the background modes Ψ_2, Ψ_4 , and Ψ_6 ,
penev_B_F00267	8	0.993	Ψ_1^{12}	Ψ_1^{12}	will suffice—a very frugal definition of Ψ_1^{12}
penev_B_F00268	8	1.000	$\alpha, \mathcal{P}_\alpha$	$\alpha, \mathcal{P}_\alpha$	$\mathcal{P}_{objects}$, and the symmetric density of the parameter of the breaking $\alpha, \mathcal{P}_\alpha$.
penev_B_F00269	8	0.988	β_{feature}	β_{feature}	$\beta_{feature}$, as a parameter for the breaking of the local symmetry. Analogously,
penev_B_F00270	8	0.994	$\mathcal{P}_{\text{feature}}$	$\mathcal{P}_{\text{feature}}$	ensemble, this time of features, and also, its probability density $\mathcal{P}_{feature}$, along
penev_B_F00271	8	1.000	$\mathcal{P}_\beta$	$\mathcal{P}_\beta$	with the probability density $\mathcal{P}_\beta$, as a parameterization of $\mathcal{P}_{objects}$.
penev_B_F00272	8	1.000	Ψ_r	Ψ_r	at the fovea; this obviates the need for any genetic storage of Ψ_r .
penev_B_F00273	8	0.429	$\phi^t(\mathbf{x})$	$\phi^t(\mathbf{x})$	receptors”— $\phi^t(\mathbf{x})$ was interpreted as the luminosity value at the grid point $\mathbf{x}$. This worked
penev_B_F00274	8	1.000	A	A	weak assumptions, if area A projects forward to area B , then almost always
penev_B_F00275	8	1.000	B	B	weak assumptions, if area A projects forward to area B , then almost always
penev_B_F00276	8	0.932	20%	20%	Ensemble 6, where about 20% of the faces have eyeglasses, the “cheek-bone”
penev_B_F00277	8	0.996	$P_m(\mathbf{x})$	$P_m(\mathbf{x})$	Unfortunately, its reconstructor $P_m(\mathbf{x})$ (Fig. 5.4e) will tend to create an illusion
penev_B_F00278	8	0.662	$\phi_{100}^{\text{err}} - \phi_{500}^{\text{err}}$	$\phi_{100}^{\text{err}} - \phi_{500}^{\text{err}}$	of faint eyeglasses (cf. $\phi_{100}^{\text{err}} - \phi_{500}^{\text{err}}$ on Fig. 5.2), ¹⁵ and it will take many more
penev_B_F00279	8	0.919	$\left\{ \alpha_{G.M.}^t \right\}_{t \in T}$	$\{ \alpha_{G.M.}^t \}_{t \in T}$	well for two reasons: we had a good initial guess $\{ \alpha_{G.M.}^t \}_{t \in T}$ about the symmetry breaking
penev_B_F00280	8	0.769	$\mathcal{P}_{\text{abjects}}$	$\mathcal{P}_{\text{abjects}}$	$\mathcal{P}_{objects}$, probability distributions in one representational module (Fig. 5.1). In general, when
penev_B_F00281	8	0.916	A_3	A_3	the forward, or <i>ascending</i> , pathway originate in the superficial pyramidal cells of area A ,
penev_B_F00282	9	0.985	$O^{\text{err}}(\mathbf{x})$	$O^{\text{err}}(\mathbf{x})$	ule will then calculate $O^{\text{err}}(\mathbf{x})$ and will feed it to the cheekbone module for
penev_B_F00283	9	1.000	U	U	achieved by topography and the simplistic fixing of the U -symmetry (4.16); as long as we
penev_B_F00284	9	0.996	$\mathcal{M} \subset V$	$\mathcal{M} \subset V$	values in a sparse subset $\mathcal{M} \subset V$ of the receptor grid $\{ \mathbf{x} \}_{\mathbf{x} \in V}$ (Fig. 4.6). An
penev_B_F00285	9	0.996	$\{ \mathbf{x} \}_{\mathbf{x} \in V}$	$\{ \mathbf{x} \}_{\mathbf{x} \in V}$	values in a sparse subset $\mathcal{M} \subset V$ of the receptor grid $\{ \mathbf{x} \}_{\mathbf{x} \in V}$ (Fig. 4.6). An
penev_B_F00286	9	0.868	$\mathbf{P} = \mathbf{O}^* \mathbf{O} (4.28)$	$\mathbf{P} = \mathbf{O}^* \mathbf{O} (4.28)$	residual correlations $\mathbf{P} = \mathbf{O}^* \mathbf{O} (4.28)$, and, in the symmetric case, their support
penev_B_F00287	9	0.949	$\mathbf{O} * \mathbf{O} = 1 (4.12)$	$\mathbf{O} * \mathbf{O} = 1 (4.12)$	collapses to the point in the center, $\mathbf{O}^* \mathbf{O} = \mathbf{1} (4.12)$. Therefore, the knowledge
penev_B_F00288	9	0.536	$\left\{ O(\mathbf{x}_m) \right\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{ O(\mathbf{x}_m) \}_{\mathbf{x}_m \in \mathcal{M}}$	of the values $\{ O(\mathbf{x}_m) \}_{\mathbf{x}_m \in \mathcal{M}}$ does not provide any information about the rest
penev_B_F00289	9	0.961	$\mathbf{f} \in \Delta_j$	$\mathbf{f} \in \Delta_j$	$\mathbf{f} \in \Delta_j$. Therefore, the correlation function of the sensory ensemble Φ_j , as seen
penev_B_F00290	9	0.961	Φ_j	Φ_j	$\mathbf{f} \in \Delta_j$. Therefore, the correlation function of the sensory ensemble Φ_j , as seen
penev_B_F00291	9	0.994	j	j	by a neuron in the j -th frequency band, is $\mathbf{R}_j \equiv \mathbf{K}_j^{(-2)}$ with (cf. 4.27)
penev_B_F00292	9	0.994	$\mathbf{R}_j \equiv \mathbf{K}_j^{(-2)}$	$\mathbf{R}_j \equiv \mathbf{K}_j^{(-2)}$	by a neuron in the j -th frequency band, is $\mathbf{R}_j \equiv \mathbf{K}_j^{(-2)}$ with (cf. 4.27)
penev_B_F00293	9	0.692	$\phi_j = \mathbf{P}_j \phi$	$\phi_j = \mathbf{P}_j \phi$	The sensory signal that a multi-scale neuron sees is $\phi_j = \mathbf{P}_j \phi$ with $\mathbf{P}_j = \mathbf{K}_j^{(0)}$,
penev_B_F00294	9	0.692	$\mathbf{P}_j = \mathbf{K}_j^{(0)}$	$\mathbf{P}_j = \mathbf{K}_j^{(0)}$	The sensory signal that a multi-scale neuron sees is $\phi_j = \mathbf{P}_j \phi$ with $\mathbf{P}_j = \mathbf{K}_j^{(0)}$,
penev_B_F00295	9	0.910	$\mathbf{K}_j = \mathbf{K}_j^{(1)}$	$\mathbf{K}_j = \mathbf{K}_j^{(1)}$	Now, the outputs of the action of $\mathbf{K}_j = \mathbf{K}_j^{(1)}$ on ϕ_j is ²⁰
penev_B_F00296	9	0.910	ϕ_j	ϕ_j	Now, the outputs of the action of $\mathbf{K}_j = \mathbf{K}_j^{(1)}$ on ϕ_j is ²⁰
penev_B_F00297	9	0.499	$O(\mathbf{x})$	$O(\mathbf{x})$	²⁰ Notably, j is not a substitute for the receptor grid index $\mathbf{x}$ in $O(\mathbf{x})$; there are several band-
penev_B_F00298	9	1.000	O_j	O_j	passed dense LFA outputs O_j , which collectively carry the information that was previously
penev_B_F00299	9	1.000	O	O	carried entirely by O . When we want to emphasize their dependence on $\{ \mathbf{x} \}$, we will write
penev_B_F00300	9	1.000	$\{ \mathbf{x} \}$	$\{ \mathbf{x} \}$	carried entirely by O . When we want to emphasize their dependence on $\{ \mathbf{x} \}$, we will write
penev_B_F00301	9	0.439	$\left(\hat{\mathbf{x}}, O_j \right) = O_j(\mathbf{x})$	$(\hat{\mathbf{x}}, O_j) = O_j(\mathbf{x})$	$(\hat{\mathbf{x}}, O_j) = O_j(\mathbf{x})$. For the ensemble of monaural sound, $\mathbf{x}$ is the time index t ; for the static
penev_B_F00302	9	0.439	t	t	$(\hat{\mathbf{x}}, O_j) = O_j(\mathbf{x})$. For the ensemble of monaural sound, $\mathbf{x}$ is the time index t ; for the static

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00303	9	■ 0.999	(x, y)	(x, y)	natural images, the pair (x, y) ; for the dynamic natural “movies,” the triplet (x, y, t) .
penev_B_F00304	9	■ 0.999	(x, y, t)	(x, y, t)	natural images, the pair (x, y) ; for the dynamic natural “movies,” the triplet (x, y, t) .
penev_B_F00305	9	■ 1.000	$\left\{\Delta_j\right\}$	$\{\Delta_j\}$	²¹ Here, we are not addressing the possibility that the bands $\{\Delta_j\}$ may happen to be non-
penev_B_F00306	9	■ 0.999	$\mathrm{O}_{\{j\}^*} \mathrm{O}_{\{k\}}$	$O_j^* O_k$	question what $O_j^* O_k$ is for $j \neq k$.
penev_B_F00307	9	■ 0.999	$j \neq k$	$j \neq k$	question what $O_j^* O_k$ is for $j \neq k$.
penev_B_F00308	9	■ 1.000	$F_{\{r\}}$	F_r	One way to deal uniformly with all cases is through the “noise suppression” factor F_r .
penev_B_F00309	9	■ 0.995	$(\mathbf{P}=\mathbf{1})$	$(\mathbf{P} = \mathbf{1})$	page 61); it could be decorrelated fully $(\mathbf{P} = \mathbf{1})$ by the retinal filters (4.26), but
penev_B_F00310	9	■ 1.000	$\left\{\phi_j\right\}$	$\{\phi_j\}$	among the bands $\{\Delta_j\}$, resulting in $\{\phi_j\}$, so is the dense LFA output, $\{O_j\}$.
penev_B_F00311	9	■ 1.000	$\left\{O_j\right\}$	$\{O_j\}$	among the bands $\{\Delta_j\}$, resulting in $\{\phi_j\}$, so is the dense LFA output, $\{O_j\}$.
penev_B_F00312	9	■ 1.000	$\mathrm{O}_{\{j\}}$	O_j	the respective output ensembles O_j , are symmetric. The multi-scale process-
penev_B_F00313	9	■ 1.000	$O_{\{3\}}$	O_3	after a specific example ϕ has been presented and the degenerate outputs O_j
penev_B_F00314	9	■ 0.858	$\mathbf{K}_{\{j\}^n}(\mathbf{n})=\sum_{r \in \Delta_j} \psi_r\left(\frac{1}{\sigma_r}\right)^n\left(\psi_r,\right) \psi_i\left(\frac{1}{\sigma_i}\right)^n\left(\psi_i,\right)$	$\mathbf{K}_j^{(\mathbf{n})}=\sum_{r \in \Delta_j} \psi_r\left(\frac{1}{\sigma_r}\right)^n\left(\psi_r,\right)$	representation of the type $\mathbf{K}_j^{(\mathbf{n})}=\sum_{r \in \Delta_j} \psi_r\left(\frac{1}{\sigma_r}\right)^n\left(\psi_r,\right)$ (cf. 4.30) can result in additional
penev_B_F00315	9	■ 0.695	$\left\{\mathrm{As}\right\}$	$^{23}\mathrm{As}$	$^{23}\mathrm{As}$ with any symmetry breaking, it can be understood as a definition of a restricted
penev_B_F00316	10	■ 0.723	$\left\{a_m(\mathbf{x})\right\}_{x_m \in \mathcal{M}}(4.68)$	$\{a_m(\mathbf{x})\}_{x_m \in \mathcal{M}}(4.68)$	struction as either the set of interpolation reconstructors $\{a_m(\mathbf{x})\}_{x_m \in \mathcal{M}}(4.68)$
penev_B_F00317	10	■ 0.869	$P_{\{m\}}(\mathbf{x})(4.73)$	$P_m(\mathbf{x})(4.73)$	$P_m(\mathbf{x})(4.73)$.
penev_B_F00318	10	■ 1.000	$ \mathcal{M} =M-1$	$ \mathcal{M} =M-1$	We will consider the case when $ \mathcal{M} =M-1$ units are currently active and the
penev_B_F00319	10	■ 1.000	M	M	M -th one is being activated. ²⁷ Then, two things happen in the reconstruction
penev_B_F00320	10	■ 1.000	$M-1$	$M-1$	generated; and corrections to the old $(M-1)$ reconstructors are made.
penev_B_F00321	10	■ 0.403	$\mathbf{Q}(4.63)$	$\mathbf{Q}(4.63)$	inversion of the matrix $\mathbf{Q}(4.63)$ and the correlators $P_m(\mathbf{x})(4.68)$; the latter
penev_B_F00322	10	■ 0.403	$P_{\{m\}}(\mathbf{x})(4.68)$	$P_m(\mathbf{x})(4.68)$	inversion of the matrix $\mathbf{Q}(4.63)$ and the correlators $P_m(\mathbf{x})(4.68)$; the latter
penev_B_F00323	10	■ 0.999	$n=n^{\prime}+n^{\prime \prime}$	$n=n^{\prime}+n^{\prime \prime}$	In general, for $n=n^{\prime}+n^{\prime \prime}$, any $n \times n$ matrix W can be partitioned as
penev_B_F00324	10	■ 0.999	$n \times n$	$n \times n$	In general, for $n=n^{\prime}+n^{\prime \prime}$, any $n \times n$ matrix W can be partitioned as
penev_B_F00325	10	■ 0.999	W	W	In general, for $n=n^{\prime}+n^{\prime \prime}$, any $n \times n$ matrix W can be partitioned as
penev_B_F00326	10	■ 0.745	$\mathbf{1}_{n^{\prime}}$	$\mathbf{1}_{n^{\prime}}$	where $\mathbf{1}_{n^{\prime}}$ is the $n^{\prime} \times n^{\prime}$ unit matrix; analogously for $\mathbf{1}_{n^{\prime \prime}}$; and the partition-
penev_B_F00327	10	■ 0.745	$n^{\prime} \times n^{\prime}$	$n^{\prime} \times n^{\prime}$	where $\mathbf{1}_{n^{\prime}}$ is the $n^{\prime} \times n^{\prime}$ unit matrix; analogously for $\mathbf{1}_{n^{\prime \prime}}$; and the partition-
penev_B_F00328	10	■ 0.745	$\mathbf{1}_{n^{\prime \prime}}$	$\mathbf{1}_{n^{\prime \prime}}$	where $\mathbf{1}_{n^{\prime}}$ is the $n^{\prime} \times n^{\prime}$ unit matrix; analogously for $\mathbf{1}_{n^{\prime \prime}}$; and the partition-
penev_B_F00329	10	■ 1.000	W^{-1}	W^{-1}	derstanding that they are as in (5.9). Then, for the inverse W^{-1} , we have
penev_B_F00330	10	■ 0.999	w^{-1}	w^{-1}	Thus, W^{-1} is calculated by a correction to w^{-1} , and only the quantities $x, y,$
penev_B_F00331	10	■ 0.999	x, y	x, y	Thus, W^{-1} is calculated by a correction to w^{-1} , and only the quantities $x, y,$
penev_B_F00332	10	■ 1.000	z	z	and z are needed additionally; W and w are never needed explicitly.
penev_B_F00333	10	■ 1.000	w	w	and z are needed additionally; W and w are never needed explicitly.
penev_B_F00334	10	■ 1.000	$M \times M$	$M \times M$	After the M -th unit becomes active, we identify W with the $M \times M$ ma-
penev_B_F00335	10	■ 0.513	$(n=M)$	$(n=M)$	trix $\mathbf{Q}(n=M)$, whose matrix elements are (4.63)
penev_B_F00336	10	■ 0.933	$(M-1) \times (M-1)$	$(M-1) \times (M-1)$	Also, we identify w with the $(M-1) \times (M-1)$ matrix from the previous step,
penev_B_F00337	10	■ 0.861	$w \equiv q$	$w \equiv q$	$w \equiv q$, comprised by the first $M-1$ rows and columns of $\mathbf{Q}, q_{ml}=Q_{ml} _{l, m=1}^{M-1}$.
penev_B_F00338	10	■ 0.861	$\mathbf{Q}, q_{ml}=\left.Q_{ml}\right _{l, n s=1}^{M-1}$	$\mathbf{Q}, q_{ml}=Q_{ml} _{l, n s=1}^{M-1}$	$w \equiv q$, comprised by the first $M-1$ rows and columns of $\mathbf{Q}, q_{ml}=Q_{ml} _{l, m=1}^{M-1}$.
penev_B_F00339	10	■ 0.994	$w, n^{\prime}=M-1$	$w, n^{\prime}=M-1$	With these W and $w, n^{\prime}=M-1$, hence $n^{\prime \prime}=1$; then x is a $(M-1)$ -
penev_B_F00340	10	■ 0.994	$n^{\prime \prime}=1$	$n^{\prime \prime}=1$	With these W and $w, n^{\prime}=M-1$, hence $n^{\prime \prime}=1$; then x is a $(M-1)$ -
penev_B_F00341	10	■ 0.994	x	x	With these W and $w, n^{\prime}=M-1$, hence $n^{\prime \prime}=1$; then x is a $(M-1)$ -
penev_B_F00342	10	■ 0.994	$(M-1)$	$(M-1)$	With these W and $w, n^{\prime}=M-1$, hence $n^{\prime \prime}=1$; then x is a $(M-1)$ -
penev_B_F00343	10	■ 0.908	y	y	dimensional column vector; y , a row vector; z and $\bar{z}$, scalars. Because $\mathbf{P}(4.29)$
penev_B_F00344	10	■ 0.908	$\bar{z}$	$\bar{z}$	dimensional column vector; y , a row vector; z and $\bar{z}$, scalars. Because $\mathbf{P}(4.29)$
penev_B_F00345	10	■ 0.806	p	p	and, therefore, $\mathbf{Q}$ are symmetric, we can define p as the row vector ²⁸
penev_B_F00346	10	■ 0.917	$\tilde{z}$	$\tilde{z}$	Because $\mathbf{Q}$ is symmetric and $\tilde{z}$ is a scalar, the second term in (5.11) factorizes
penev_B_F00347	10	■ 0.870	$O^{r e e}(\mathbf{x})$	$O^{r e e}(\mathbf{x})$	means that $O^{r e c}(\mathbf{x})$ is a V -dimensional column vector, which is a linear com-
penev_B_F00348	10	■ 1.000	$A^m(\mathbf{x})$	$A^m(\mathbf{x})$	bination of the M (uppercase) column vectors $A^m(\mathbf{x})$ with the M coefficients
penev_B_F00349	10	■ 1.000	$O_m=O\left(\mathbf{x}_m\right)$	$O_m=O\left(\mathbf{x}_m\right)$	$O_m=O\left(\mathbf{x}_m\right)$. Analogously, the column vector $A^m(\mathbf{x})$ for the m -th construc-
penev_B_F00350	10	■ 1.000	$^{28} p^T$	$^{28} p^T$	$^{28} p^T$ is a column vector, which is the <i>transpose</i> of the row vector p .
penev_B_F00351	10	■ 1.000	$P^l(\mathbf{x})$	$P^l(\mathbf{x})$	is a linear combination of the M correlators $P^l(\mathbf{x})$, centered at the active grid
penev_B_F00352	10	■ 0.967	$\mathbf{x}_l$	$\mathbf{x}_l$	points, $\mathbf{x}_l$, with the coefficients $(\mathbf{Q}^{-1})_l^m$, from the m -th column of $\mathbf{Q}^{-1}$.
penev_B_F00353	10	■ 0.967	$\left(\mathbf{Q}^{-1}\right)_l^m$	$(\mathbf{Q}^{-1})_l^m$	points, $\mathbf{x}_l$, with the coefficients $(\mathbf{Q}^{-1})_l^m$, from the m -th column of $\mathbf{Q}^{-1}$.
penev_B_F00354	10	■ 0.994	$\left[a^m(\mathbf{x}) \quad 0\right]$	$[a^m(\mathbf{x}) \quad 0]$	Thus, the new reconstructors $A^m(\mathbf{x})$ are the old ones, $[a^m(\mathbf{x}) \quad 0]$, with a
penev_B_F00355	10	■ 0.880	$A(\mathbf{x})$	$A(\mathbf{x})$	whose entire $\mathbf{x}$ -dependence is given by $A(\mathbf{x})$. Indeed, the new reconstructor

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00356	10	■ 1.000	$A^{\{M\}}(\mathbf{x})$	$A^M(\mathbf{x})$	$A^M(\mathbf{x})$ is proportional to $A(\mathbf{x})$ with a coefficient $\bar{z}$,
penev_B_F00357	10	■ 1.000	$\tilde{z}_1$	$\tilde{z}_1$	$A^M(\mathbf{x})$ is proportional to $A(\mathbf{x})$ with a coefficient $\bar{z}$,
penev_B_F00358	10	■ 0.886	$a^{\{m\}}(\mathbf{x})$	$a^m(\mathbf{x})$	and the correction to any old reconstructor $a^m(\mathbf{x})$, is proportional to $A^M(\mathbf{x})$
penev_B_F00359	10	■ 0.886	$A^{\{M\}}(\mathbf{x})$	$A^M(\mathbf{x})$	and the correction to any old reconstructor $a^m(\mathbf{x})$, is proportional to $A^M(\mathbf{x})$
penev_B_F00360	10	■ 1.000	$-u^{\{m\}}$	$-u^m$	with a coefficient $-u^m$:
penev_B_F00361	10	■ 0.981	$u=p\ q^{\{1\}}$	$u=pq^1$	Now, we are well motivated to inquire into the meaning of $u=pq^{-1}$ (5.15).
penev_B_F00362	10	■ 0.993	$u^{\{m\}}$	u^m	From (5.13) (cf. 5.18), u^m is the old reconstructor $a^m(\mathbf{x})$, evaluated at the
penev_B_F00363	10	■ 0.679	$\mathbf{x}_M$	$\mathbf{x}_M$	newly active point, $\mathbf{x}_M$,
penev_B_F00364	10	■ 0.815	$p^{\{l\}}(\mathbf{x})\ u^{\{T_l\}}=p^{\{l\}}(\mathbf{x})\left(q^{-1}\right)_l^m P^M\left(\mathbf{x}_m\right)=a^m(\mathbf{x}) P^M\left(\mathbf{x}_m\right)$	$p^l(\mathbf{x})u^{T_l}=p^l(\mathbf{x})(q^{-1})_l^m P^M(\mathbf{x}_m)=a^m(\mathbf{x})P^M(\mathbf{x}_m)$	Also, (cf. 5.20), $p^l(\mathbf{x})u^{T_l}=p^l(\mathbf{x})(q^{-1})_l^m P^M(\mathbf{x}_m)=a^m(\mathbf{x})P^M(\mathbf{x}_m)$; hence
penev_B_F00365	10	■ 0.943	$a^{\{m\}}(\mathbf{x})\ P^{\{M\}}\left(\mathbf{x}_m\right)$	$a^m(\mathbf{x})P^M(\mathbf{x}_m)$	The last term in (5.25), $a^m(\mathbf{x})P^M(\mathbf{x}_m)$, is analogous to (5.17) with the
penev_B_F00366	10	■ 0.629	$\left\{\mathbf{x}_m\right\}$	$\{\mathbf{x}_m\}$	evaluated at $\{\mathbf{x}_m\}$ is $P^M(\mathbf{x})$, rather than $O(\mathbf{x})$. Therefore, it is useful, given
penev_B_F00367	10	■ 0.629	$P^{\{M\}}(\mathbf{x})$	$P^M(\mathbf{x})$	evaluated at $\{\mathbf{x}_m\}$ is $P^M(\mathbf{x})$, rather than $O(\mathbf{x})$. Therefore, it is useful, given
penev_B_F00368	10	■ 1.000	$V(\mathbf{x})$	$V(\mathbf{x})$	the current restriction $\mathcal{M}$, to define, for any vector $V(\mathbf{x})$, the <i>reconstruction</i>
penev_B_F00369	11	■ 0.926	$P^{\{M\}}(\mathbf{x})$	$P^M(\mathbf{x})$	Remarkably, $A(\mathbf{x})$ is the difference between the true value of $P^M(\mathbf{x})$ and its
penev_B_F00370	11	■ 1.000	$\bar{z}^{-1}=\bar{z}-u p^T$	$\bar{z}^{-1}=\bar{z}-u p^T$	$\bar{z}^{-1}=\bar{z}-u p^T$, and from (5.14), (5.24), (5.13), and (5.29),
penev_B_F00371	11	■ 0.868	$a^{\{m\}}(\mathbf{x})$	$a^m(\mathbf{x})$	predictors, $a^m(\mathbf{x})$, are corrected by a negative feedback of this error, in propor-
penev_B_F00372	11	■ 0.870	$a^{\{m\}}\left(\mathbf{x}_M\right)$	$a^m(\mathbf{x}_M)$	tionality with their original prediction strengths, $a^m(\mathbf{x}_M)$.
penev_B_F00373	11	■ 0.582	$O^{\{\text {rec }\}}(\mathbf{x})=\mathcal{R}_{O}^{M-1}(\mathbf{x})$	$O^{\text{rec}}(\mathbf{x})=\mathcal{R}_O^{M-1}(\mathbf{x})$	the current estimate $O^{\text{rec}}(\mathbf{x})=\mathcal{R}_O^{M-1}(\mathbf{x})$, the current error $O^{\text{err}}(\mathbf{x})=\mathcal{D}_O(\mathbf{x})$
penev_B_F00374	11	■ 0.582	$O^{\{\text {err }\}}(\mathbf{x})=\mathcal{D}_O(\mathbf{x})$	$O^{\text{err}}(\mathbf{x})=\mathcal{D}_O(\mathbf{x})$	the current estimate $O^{\text{rec}}(\mathbf{x})=\mathcal{R}_O^{M-1}(\mathbf{x})$, the current error $O^{\text{err}}(\mathbf{x})=\mathcal{D}_O(\mathbf{x})$
penev_B_F00375	11	■ 0.381	$\Delta O^{\{\text {rec }\}}(\mathbf{x})$	$\Delta O^{\text{rec}}(\mathbf{x})$	introduces a correction $\Delta O^{\text{rec}}(\mathbf{x})$, which is exactly equal to the error at that
penev_B_F00376	11	■ 0.716	$\mathcal{D}_O\left(\mathbf{x}_M\right)$	$\mathcal{D}_O(\mathbf{x}_M)$	on the current reconstruction error, $\mathcal{D}_O(\mathbf{x}_M)$.
penev_B_F00377	11	■ 0.490	$O^{\{\text {err }\}}\left(\mathbf{x}_M\right)=\mathcal{D}_O\left(\mathbf{x}_M\right)$	$O^{\text{err}}(\mathbf{x}_M)=\mathcal{D}_O(\mathbf{x}_M)$	$O^{\text{err}}(\mathbf{x}_M)=\mathcal{D}_O(\mathbf{x}_M)$.
penev_B_F00378	11	■ 1.000	$f(t)$	$f(t)$	like to understand it as an infinite source $f(t)$ (cf. Appendix A.6), from which
penev_B_F00379	11	■ 0.551	$\phi^{\left\{t_n\right\}}\left(\mathbf{x}_k\right)=f\left(t_{\mathbf{n}-k}\right)$	$\phi^{t_n}(\mathbf{x}_k)=f(t_{\mathbf{n}-k})$	we will construct finite-dimensional ensembles of examples $\phi^{t_n}(\mathbf{x}_k)=f(t_{n-k})$
penev_B_F00380	11	■ 1.000	t_n	t_n	by imposing a finite discrete window of size V around the time t_n (A.13).
penev_B_F00381	11	■ 0.975	$\left\{f_{\chi}\right\}_{\chi \in X}$	$\{f_{\chi}\}_{\chi \in X}$	The <i>Encoder</i> (a) has access to the signal $\{f_{\chi}\}_{\chi \in X}$. Based on the part of the
penev_B_F00382	11	■ 0.467	$\mathbf{x}_n$	$\mathbf{x}_n$	location $\mathbf{x}_n$, an estimate $f^{\text{rec}}(\mathbf{x}_n) \equiv f_n^{\text{rec}}$, and the <i>Quantizer</i> quantizes it to $\hat{f}_n$.
penev_B_F00383	11	■ 0.467	$f^{\{\text {rec }\}}\left(\mathbf{x}_n\right) \equiv f_n^{\text {rec }}$	$f^{\text{rec}}(\mathbf{x}_n) \equiv f_n^{\text{rec}}$	location $\mathbf{x}_n$, an estimate $f^{\text{rec}}(\mathbf{x}_n) \equiv f_n^{\text{rec}}$, and the <i>Quantizer</i> quantizes it to $\hat{f}_n$.
penev_B_F00384	11	■ 0.467	$\hat{f}_n$	$\hat{f}_n$	location $\mathbf{x}_n$, an estimate $f^{\text{rec}}(\mathbf{x}_n) \equiv f_n^{\text{rec}}$, and the <i>Quantizer</i> quantizes it to $\hat{f}_n$.
penev_B_F00385	11	■ 0.996	$e_n=f_n-\hat{f}_n$	$e_n=f_n-\hat{f}_n$	The error $e_n=f_n-\hat{f}_n$ is calculated and is fed to the <i>Symbol Encoder</i> , which
penev_B_F00386	11	■ 1.000	e_n	e_n	error signal e_n . This error is added to the current reconstruction $\hat{f}_n$, which
penev_B_F00387	11	■ 1.000	$\hat{f}_n=\hat{f}_n+e_n$	$\hat{f}_n=\hat{f}_n+e_n$	The result, $\hat{f}_n=\hat{f}_n+e_n$, is an identical copy of $\hat{f}_n$.
penev_B_F00388	11	■ 0.934	$f^{\{\text {rec }\}}\left(t_n\right)=\hat{f}_n$	$f^{\text{rec}}(t_n)=\hat{f}_n$	mates of the of the current value, $f^{\text{rec}}(t_n)=\hat{f}_n$. The encoder, which has access
penev_B_F00389	11	■ 0.824	$e_n=f_n-\hat{f}_n$	$e_n=f_n-\hat{f}_n$	to the true value f_n , calculates the error $e_n=f_n-\hat{f}_n$ and feeds it, via the com-
penev_B_F00390	11	■ 0.999	$t_n,\left\{f_n(m)=f\left(t_{n-m}\right)=f_{n-m}\right\}_{m=1}^M$	$t_n,\{f_n(m)=f(t_{n-m})=f_{n-m}\}_{m=1}^M$	only a small time window, <i>preceding</i> t_n , $\{f_n(m)=f(t_{n-m})=f_{n-m}\}_{m=1}^M$. The
penev_B_F00391	11	■ 0.995	$\hat{f}_n=\alpha^m f_{n-m}$	$\hat{f}_n=\alpha^m f_{n-m}$	similar to (5.17)— $\hat{f}_n=\alpha^m f_{n-m}$.
penev_B_F00392	11	■ 1.000	α_m	α_m	There are many possible choices for the coefficients α_m ; when they are
penev_B_F00393	11	■ 0.998	$E\left\{e_n^2\right\}$	$E\left\{e_n^2\right\}$	chosen to minimize the expectation of the square of the error signal, $E\{e_n^2\}$, the
penev_B_F00394	11	■ 1.000	$\left\{t_{n-m}\right\}_{m=1}^{M-1}$	$\{t_{n-m}\}_{m=1}^{M-1}$	is known, $\{t_{n-m}\}_{m=1}^{M-1}$, with $\{\mathbf{x}_m\}_{m=1}^{M-1}$, and the currently reconstructed grid
penev_B_F00395	11	■ 1.000	$\left\{\mathbf{x}_m\right\}_{m=1}^{M-1}$	$\{\mathbf{x}_m\}_{m=1}^{M-1}$	is known, $\{t_{n-m}\}_{m=1}^{M-1}$, with $\{\mathbf{x}_m\}_{m=1}^{M-1}$, and the currently reconstructed grid
penev_B_F00396	11	■ 0.964	$\mathbf{x}_M$	$\mathbf{x}_M$	point t_n , with $\mathbf{x}_M$. Then $\mathbf{Q}$ (5.40) is the correlation matrix of the signal,
penev_B_F00397	12	■ 1.000	Φ	Φ	of experimentally well-controlled sensory stimuli Φ , interpreted as the horizon-
penev_B_F00398	12	■ 1.000	$v(t)$	$v(t)$	tal velocity $v(t)$; simultaneously, the spike train $\{t_m\}$ that results from this
penev_B_F00399	12	■ 1.000	$\left\{t_m\right\}$	$\{t_m\}$	tal velocity $v(t)$; simultaneously, the spike train $\{t_m\}$ that results from this
penev_B_F00400	12	■ 0.702	$A^{\{M\}}(t)=F_1\left(t-t_M\right)$	$A^M(t)=F_1(t-t_M)$	enforced, $A^M(t)=F_1(t-t_M)$, and the m.s.e., $E\{(v(t)-v^{\text{rec}}(t))^2\}$, is minimized
penev_B_F00401	12	■ 0.702	$E\left\{\left(v(t)-v^{\{\text {rec }\}}(t)\right)^2\right\}$	$E\left\{(v(t)-v^{\text{rec}}(t))^2\right\}$	enforced, $A^M(t)=F_1(t-t_M)$, and the m.s.e., $E\{(v(t)-v^{\text{rec}}(t))^2\}$, is minimized
penev_B_F00402	12	■ 1.000	$F(\tau)$	$F(\tau)$	over the experimental data set, the best $F(\tau)$ is determined. It is used to
penev_B_F00403	12	■ 0.946	$v^{\{\text {rec }\}}(t)$	$v^{\text{rec}}(t)$	calculate the mutual information between $v(t)$ and $v^{\text{rec}}(t)$, which is a lower
penev_B_F00404	12	■ 1.000	$A^{\{m\}}(t)$	$A^m(t)$	lar to (5.36), there are two major differences. The reconstructors $A^m(t)$ are

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00405	12	■ 0.424	$t_{\{M\}}.^{\{36\}}$	$t_M.^{36}$	identical—they are based on a single grid point, t_M . ³⁶
penev_B_F00406	12	■ 0.974	$100\ \%.^{\{37\}}$	$100\%.$ ³⁷	about the actual signal will be much higher, close to 100%. ³⁷
penev_B_F00407	12	■ 0.974	$\{ \}^{\{36\}}\ \mathrm{\{An\}}$	$^{36}\mathrm{An}$	³⁶ An attempt to base them on two grid points, t_M and t_{M-1} , albeit also in a time-
penev_B_F00408	12	■ 0.974	$t_{\{M\}}$	t_M	³⁶ An attempt to base them on two grid points, t_M and t_{M-1} , albeit also in a time-
penev_B_F00409	12	■ 0.974	$t_{\{M-1\}}$	t_{M-1}	³⁶ An attempt to base them on two grid points, t_M and t_{M-1} , albeit also in a time-
penev_B_F00410	12	■ 0.998	$5-15\ \%$	$5-15\%$	(5-15%, depending on the system under study).
penev_B_F00411	12	■ 0.996	$\mathrm{\mathcal{P}}_{\{\mathrm{\text{objects}}\}}$	$\mathcal{P}_{\mathrm{objects}}$	of the probability densities of images, $\mathcal{P}_{\mathrm{images}}$, and of objects, $\mathcal{P}_{\mathrm{objects}}$. We
penev_B_F00412	12	■ 0.964	$\mathrm{\mathcal{P}}_{\{\alpha\}}$	$\mathcal{P}_{\alpha}$	understood the probability distributions $\mathcal{P}_{\mathcal{O}}$ and $\mathcal{P}_{\mathrm{objects}}$ as part of the explicit
penev_B_F00413	12	■ 0.515	$\mathrm{\mathcal{P}}_{\{\mathrm{\text{imnges}}\}}$	$\mathcal{P}_{\mathrm{imnges}}$	parameterization of $\mathcal{P}_{\mathrm{images}}$.
penev_B_F00414	12	■ 0.800	$\mathrm{\mathcal{P}}_{\{\mathrm{o\ b\ j\ e\ c\ t\ s}\}}$	$\mathcal{P}_{\mathrm{objects}}$	the re-parameterization of $\mathcal{P}_{\mathrm{objects}}$ with $\mathcal{P}_{\beta}$ and $\mathcal{P}_{\mathrm{feature}}$. This hierarchical pa-
penev_B_F00415	13	■ 0.989	$O^{\{\mathrm{r\ e\ c}\}}(\mathrm{\mathbf{x}})=\mathrm{\mathcal{R}}_{\{O\}}^{\{M-1\}}(\mathrm{\mathbf{x}})$	$O^{\mathrm{rec}}(\mathbf{x}) = \mathcal{R}_O^{M-1}(\mathbf{x})$	identical reconstructions $O^{\mathrm{rec}}(\mathbf{x}) = \mathcal{R}_O^{M-1}(\mathbf{x})$; the encoder calculates the recon-
penev_B_F00416	13	■ 0.991	$\mathrm{\mathcal{D}}_{\{\mathrm{\mathcal{D}}_{\{0\}}\}}(\mathrm{\mathbf{x}})$	$\mathcal{D}_O(\mathbf{x})$	$\mathcal{D}_O(\mathbf{x})$ was largest. Here, we pose the question, Is it possible to find another
penev_B_F00417	13	■ 0.945	$\mathrm{\mathcal{D}}_{\{0\}}\left(\mathrm{\mathbf{x}}_{\{\mathrm{\text{spike}}\}}\right)=\theta$	$\mathcal{D}_O(\mathbf{x}_{\mathrm{spike}}) = \theta$	time the current reconstruction error is equal to the threshold, $\mathcal{D}_O(\mathbf{x}_{\mathrm{spike}}) = \theta$.
penev_B_F00418	13	■ 0.999	$\mathrm{\mathcal{D}}_{\{0\}}(\mathrm{\mathbf{x}})^2$	$\mathcal{D}_O(\mathbf{x})^2$	the individual grid points, $\mathcal{D}_O(\mathbf{x})^2$. On one hand, when θ is relatively large,
penev_B_F00419	13	■ 0.938	$\mathrm{\mathcal{R}}_{\{O\}}^{\{M\}}(\mathrm{\mathbf{x}})$	$\mathcal{R}_O^M(\mathbf{x})$	tion, $\mathcal{R}_O^M(\mathbf{x})$, but also, a current expectation of the variance of the error,
penev_B_F00420	13	■ 0.867	$D(\mathrm{\mathbf{x}})^2=E\left\{\mathrm{\mathcal{D}}_{\{0\}}(\mathrm{\mathbf{x}})^2\right\}$	$D(\mathbf{x})^2 = E\{\mathcal{D}_O(\mathbf{x})^2\}$	$D(\mathbf{x})^2 = E\{\mathcal{D}_O(\mathbf{x})^2\}$ (see discussion in footnote 32 on page 77). Then, θ may
penev_B_F00421	13	■ 0.495	$\theta\left(\mathrm{\mathbf{x}}-\mathrm{\mathbf{x}}_{\{\mathrm{\text{vpike}}\}}\right)$	$\theta(\mathbf{x} - \mathbf{x}_{\mathrm{vpike}})$	decays with the time since the last spike, $\theta(\mathbf{x} - \mathbf{x}_{\mathrm{vpike}})$.
penev_B_F00422	13	■ 0.508	$\mathrm{\mathcal{D}}_{\{0\}}(\mathrm{\mathbf{x}})$	$\mathcal{D}_O(\mathbf{x})$	the timing, which generates an error in the decoder's estimate of $\mathcal{D}_O(\mathbf{x})$ and,
penev_B_F00423	13	■ 0.842	$O^{\{\mathrm{r\ e\ c}\}}(\mathrm{\mathbf{x}})$	$O^{\mathrm{rec}}(\mathbf{x})$	therefore, of $O^{\mathrm{rec}}(\mathbf{x})$, which depends on the <i>derivative</i> of $O(\mathbf{x})$. ³¹ One way to
penev_B_F00424	13	■ 0.842	$O(\mathrm{\mathbf{x}}).^{\{31\}}$	$O(\mathbf{x}).$ ³¹	therefore, of $O^{\mathrm{rec}}(\mathbf{x})$, which depends on the <i>derivative</i> of $O(\mathbf{x})$. ³¹ One way to
penev_B_F00425	13	■ 1.000	Δt	Δt	is determined with precision Δt , we may consider that a spike can be triggered
penev_B_F00426	13	■ 1.000	$D(\mathrm{\mathbf{x}})$	$D(\mathbf{x})$	compares the target function, for example $D(\mathbf{x})$, with the threshold $\theta(\mathbf{x})$. Ob-
penev_B_F00427	13	■ 1.000	$\theta(\mathrm{\mathbf{x}})$	$\theta(\mathbf{x})$	compares the target function, for example $D(\mathbf{x})$, with the threshold $\theta(\mathbf{x})$. Ob-
penev_B_F00428	13	■ 0.985	$\mathrm{\mathbf{x}}_{\{\mathrm{\text{m}}\}}$	$\mathbf{x}_{\mathrm{m}}$	the difference is positive at the moment of the spike, $\mathbf{x}_{\mathrm{m}}$, but was negative the
penev_B_F00429	13	■ 0.999	$\mathrm{\mathbf{x}}^{\{\prime\}}$	$\mathbf{x}'$	preceding moment, $\mathbf{x}'$.
penev_B_F00439	13	■ 1.000	$U_{\{m\}}$	U_m	U_m and U'_m .
penev_B_F00440	13	■ 1.000	$U_{\{m\}}^{\{\prime\}}$	U'_m	U_m and U'_m .
penev_B_F00442	13	■ 0.843	$U^{\{\perp\}}$	$U^{\perp}$	All other $\tilde{N} - 2$ dimensions of the PCA S -subspace are collapsed to $U^{\perp}$.
penev_B_F00446	13	■ 1.000	C	C	to the circle, and which intersects with U'_m at point C . This hyper-plane's
penev_B_F00447	13	■ 0.914	$\phi=\Phi^{\{*\}}$	$\phi = \Phi^*$	the context of sparsification. Given the example $\phi = \Phi^*\mathbf{s}$, and the choice
penev_B_F00448	13	■ 0.380	$\mathrm{\mathbf{s}}_{\{\parallel\}}=0_{\{m\}}\ \mathrm{\mathbf{p}}_{\{\mathrm{\mathbf{x}}_{\mathrm{m}}\}}\in U_m(4.42)$	$\mathbf{s}_{\parallel} = O_m \mathbf{p}_{\mathbf{x}_{\mathrm{m}}} \in U_m(4.42)$	$\mathbf{s}_{\parallel} = O_m \mathbf{p}_{\mathbf{x}_{\mathrm{m}}} \in U_m(4.42)$, point A on Fig. 5.10, and a vector $\mathbf{s}_{\perp}$, orthogonal
penev_B_F00449	13	■ 0.380	$\mathrm{\mathbf{s}}_{\{\perp\}}$	$\mathbf{s}_{\perp}$	$\mathbf{s}_{\parallel} = O_m \mathbf{p}_{\mathbf{x}_{\mathrm{m}}} \in U_m(4.42)$, point A on Fig. 5.10, and a vector $\mathbf{s}_{\perp}$, orthogonal
penev_B_F00450	13	■ 0.829	$\mathrm{\mathbf{s}}_{\{\parallel\}}$	$\mathbf{s}_{\parallel}$	to it. We interpreted $\mathbf{s}_{\parallel}$ as the maximum likelihood estimate.
penev_B_F00451	13	■ 1.000	$O\left(\mathrm{\mathbf{x}}_m\right)=0_{\{m\}}$	$O(\mathbf{x}_m) = O_m$	decoder cannot assume that $O(\mathbf{x}_m) = O_m$ exactly, but only that $O(\mathbf{x}_m) \geq O_m$.
penev_B_F00452	13	■ 1.000	$O\left(\mathrm{\mathbf{x}}_m\right)\geq 0_{\{m\}}$	$O(\mathbf{x}_m) \geq O_m$	decoder cannot assume that $O(\mathbf{x}_m) = O_m$ exactly, but only that $O(\mathbf{x}_m) \geq O_m$.
penev_B_F00453	13	■ 0.989	$O\left(\mathrm{\mathbf{x}}^{\{\prime\}}\right)$	$O(\mathbf{x}')$	positive transition—that the value $O(\mathbf{x}')$, immediately prior to the spike, was
penev_B_F00454	13	■ 0.822	$O^{\{\prime\}}.\{ \}^{\{33\}}$	$O'.$ ³³	below a certain threshold, O' . ³³ We can decompose $\mathbf{s}$ into a sum of three mutu-
penev_B_F00455	13	■ 1.000	$\mathrm{\mathbf{s}}_{\{\parallel\}}\in U_m, \mathrm{\mathbf{s}}^{\{\prime\}}\in U_m^{\{\prime\}}$	$\mathbf{s}_{\parallel} \in U_m, \mathbf{s}' \in U'_m$	ally orthogonal vectors, $\mathbf{s}_{\parallel} \in U_m, \mathbf{s}' \in U'_m$, and $\mathbf{s}_{\perp} \in U^{\perp}$. Since both $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_m}$
penev_B_F00456	13	■ 1.000	$\mathrm{\mathbf{s}}_{\{\perp\}}\in U^{\perp}$	$\mathbf{s}_{\perp} \in U^{\perp}$	ally orthogonal vectors, $\mathbf{s}_{\parallel} \in U_m, \mathbf{s}' \in U'_m$, and $\mathbf{s}_{\perp} \in U^{\perp}$. Since both $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_m}$
penev_B_F00457	13	■ 1.000	$\mathrm{\mathbf{q}}^{\{\prime\}}$	$\mathbf{q}'$	ally orthogonal vectors, $\mathbf{s}_{\parallel} \in U_m, \mathbf{s}' \in U'_m$, and $\mathbf{s}_{\perp} \in U^{\perp}$. Since both $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_m}$
penev_B_F00458	13	■ 1.000	$\mathrm{\mathbf{q}}_{\{\mathrm{\mathbf{x}}_m\}}$	$\mathbf{q}_{\mathbf{x}_m}$	ally orthogonal vectors, $\mathbf{s}_{\parallel} \in U_m, \mathbf{s}' \in U'_m$, and $\mathbf{s}_{\perp} \in U^{\perp}$. Since both $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_m}$
penev_B_F00459	13	■ 0.999	$U^{\{\perp\}}, \mathrm{\mathbf{s}}_{\{\perp\}}$	$U^{\perp}, \mathbf{s}_{\perp}$	are orthogonal to $U^{\perp}$, $\mathbf{s}_{\perp}$ can be anywhere in $U^{\perp}$, and a maximum likelihood
penev_B_F00460	13	■ 0.972	$\mathrm{\mathbf{s}}_{\{\perp\}}=\mathrm{\mathbf{0}}$	$\mathbf{s}_{\perp} = \mathbf{0}$	estimate of it is $\mathbf{s}_{\perp} = \mathbf{0}$.
penev_B_F00461	13	■ 0.989	$\mathrm{\mathbf{s}}^{\{\prime\}}$	$\mathbf{s}'$	On the other hand, $\mathbf{s}'$ can be anywhere on the ray in U'_m , starting at C ,
penev_B_F00462	13	■ 0.998	$\mathrm{\mathbf{s}}^{\{\prime\}}=\mathrm{\mathbf{0}}$	$\mathbf{s}' = \mathbf{0}$	which is the mean of the respective distribution, is not $\mathbf{s}' = \mathbf{0}$, but is to the
penev_B_F00463	13	■ 0.994	D	D	left of A , at D . Also, the line BC intersects with U_m at some place, and that
penev_B_F00464	13	■ 0.994	BC	BC	left of A , at D . Also, the line BC intersects with U_m at some place, and that
penev_B_F00465	13	■ 0.943	$o_{\{m\}}=\left\langle \mathrm{\mathbf{q}}_{\{\mathrm{\mathbf{x}}_m\}}, \mathrm{\mathbf{s}}_{\{\parallel\}}\right\rangle$	$o_m = \langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s}_{\parallel} \rangle$	is an upper bound for $p_m = \langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s}_{\parallel} \rangle$; hence the maximum likelihood estimate
penev_B_F00466	13	■ 0.952	$o_{\{m\}}\ \mathrm{\mathbf{p}}_{\{m\}}(A)$	$O_m p_m(A)$	for $\mathbf{s}_{\parallel}$ is not $O_m p_m(A)$, but is a bit higher. ³⁴
penev_B_F00467	13	■ 0.350	$a_{\{m\}}=\left\langle \mathrm{\mathbf{q}}_{\{\mathrm{\mathbf{x}}_m\}}, \mathrm{\mathbf{s}}_{\{\parallel\}}\right\rangle$	$a_m = \langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s}_{\parallel} \rangle$	³² The values of the projection $p_m = \langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s}_{\parallel} \rangle$ will be distributed according to the orig-
penev_B_F00468	13	■ 0.966	$o_{\{m\}}\geq 0_{\{m\}}$	$o_m \geq O_m$	inal probability distribution, normal by assumption (2.49), <i>conditional</i> on $o_m \geq O_m$. The
penev_B_F00469	13	■ 0.553	$o_{\{m\}}\ \mathrm{\mathbf{p}}_{\{m\}}$	$O_m p_m$	³³ For simplicity, $O_m \mathbf{p}_m$ and $O' \mathbf{p}'$ are shown to have the same lengths on Fig. 5.10. In

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00470	13	■ 0.553	$0^{\{\backslash\mathrm{prime}\}}\mathrm{p}^{\{\backslash\mathrm{prime}\}}$	$O'p'$	³³ For simplicity, $O_m\mathbf{p}_m$ and $O'p'$ are shown to have the same lengths on Fig. 5.10. In
penev_B_F00471	13	■ 0.810	$\mathrm{q}^{\{\backslash\mathrm{prime}\}}$	q'	probability distribution along U'_m ; the smaller the angle α between $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_m}$, the more
penev_B_F00472	13	■ 0.810	$\mathrm{q}_{\{\backslash\mathrm{mathrm}\{x\}_{\backslash\mathrm{mathrm}\{m\}}\}}$	$q_{\mathbf{x}_m}$	probability distribution along U'_m ; the smaller the angle α between $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_m}$, the more
penev_B_F00473	13	■ 1.000	$C\rightarrow A$	$C\rightarrow A$	informative the lack of a spike. Ultimately, when $C\rightarrow A$, half of the axis U'_m is excluded,
penev_B_F00474	13	■ 0.848	$o_{\{m\}}$	o_m	o_m approaches O_m , and the information due to the localization of $\mathbf{s}_{ }$ in U_m increases.
penev_B_F00475	13	■ 0.848	$s_{\{\backslash \}}$	$s_{ }$	o_m approaches O_m , and the information due to the localization of $\mathbf{s}_{ }$ in U_m increases.
penev_B_F00476	13	■ 0.810	$\cos\alpha=P_{\{m\}}\backslash\mathrm{left}(x^{\{\backslash\mathrm{prime}\}}\backslash\mathrm{right})$	$\cos\alpha=P_m(x')$	This is intuitively correct; $\cos\alpha=P_m(\mathbf{x}')$, the output correlation between two adjacent
penev_B_F00477	14	■ 0.867	$\mathcal{P}_{\{\backslash\mathrm{text}\{world\}\}}$	$\mathcal{P}_{\mathrm{world}}$	parameterization of the probability density $\mathcal{P}_{\mathrm{world}}$. Using this parameterization,
penev_B_F00478	14	■ 1.000	$T=1038$	$T=1038$	We selected from the FERET database $T=1038$ examples without glasses
penev_B_F00479	14	■ 1.000	64×60	64×60	that line. Finally, the examples in the data set were cropped through a 64×60
penev_B_F00480	14	■ 0.973	$r(\backslash\mathrm{theta},\backslash\mathrm{ell})$	$r(\theta,\ell)$	ordinates $r(\theta,\ell)$ with 512 samples spanning a full revolution in the θ direction
penev_B_F00481	14	■ 1.000	$\backslash\mathrm{ell}$	ℓ	and 256 samples in the ℓ direction. Anthropological landmarks on the surfaces
penev_B_F00482	14	■ 0.823	$\backslash\mathrm{theta}_{\{\backslash\mathrm{text}\{\mathrm{fixed}\}\}},\backslash\mathrm{ell}_{\{\backslash\mathrm{text}\{\mathrm{fixed}\}\}}$	$\theta_{\mathrm{fixed}},\ell_{\mathrm{fixed}}$	The examples were aligned by us, so that the fixed point ($\theta_{\mathrm{fixed}},\ell_{\mathrm{fixed}}$)
penev_B_F00483	14	■ 0.998	3×3	3×3	smoothed with a 3×3 Gaussian filter and under-sampled twice to produce a
penev_B_F00484	14	■ 0.981	256×128	256×128	256×128 representation. All examples in the <i>Mini Survey study</i> database were
penev_B_F00485	14	■ 1.000	$T=348$	$T=348$	chosen to produce a $T=348$ data set.
penev_B_F00486	14	■ 0.999	128×64	128×64	Finally, the examples in the data set were cropped through a 128×64
penev_B_F00487	14	■ 1.000	96×90	96×90	cropped through a 96×90 window centered horizontally about the fixed point
penev_B_F00488	14	■ 1.000	96×120	96×120	cropped through a 96×120 window centered horizontally about the fixed point
penev_B_F00489	14	■ 1.000	$\backslash\mathrm{left}\{s_{\{n\}}=1.05^{\{n/2\}}\backslash\mathrm{right}\}_{\{n=-4\}}^{\{+4\}}$	$\left\{s_n=1.05^{n/2}\right\}_{n=-4}^{+4}$	ing parameter $\{s_n=1.05^{n/2}\}_{n=-4}^{+4}$, i.e., the up and down deviations from the
penev_B_F00490	14	■ 1.000	$\backslash\mathrm{approx}\{10\%,7.5\%,5\%,2.5\%,0\%\}$	$\approx\{10\%,7.5\%,5\%,2.5\%,0\%\}$	hand-determined scale were $\approx\{10\%,7.5\%,5\%,2.5\%,0\%\}$, resulting in dimen-
penev_B_F00491	14	■ 0.982	$T=10944,V=3840$	$T=10944,V=3840$	sionalities ($T=10944,V=3840$).
penev_B_F00492	14	■ 0.983	$\backslash\mathrm{left}\{s_{\{m\}}=1.025^{\{m/3\}}\backslash\mathrm{right}\}_{\{m\in\{-2,-1,1,2\}\}}$	$\left\{s_m=1.025^{m/3}\right\}_{m\in\{-2,-1,1,2\}}$	additional scalings $\{s_m=1.025^{m/3}\}_{m\in\{-2,-1,1,2\}}$, resulting in deviations of $\approx$
penev_B_F00493	14	■ 0.983	$\backslash\mathrm{approx}$	$\approx$	additional scalings $\{s_m=1.025^{m/3}\}_{m\in\{-2,-1,1,2\}}$, resulting in deviations of $\approx$
penev_B_F00494	14	■ 0.651	$V=96\times 120=11,520$	$V=96\times 120=11,520$	Three croppings were prepared—full with $V=96\times 120=11,520$, b3q
penev_B_F00495	14	■ 0.992	$V=96\times 90=8,640$	$V=96\times 90=8,640$	with $V=96\times 90=8,640$ which is the same as that for Ensemble 3, and with
penev_B_F00496	14	■ 0.881	$V=64\times 60=3,840$	$V=64\times 60=3,840$	third $V=64\times 60=3,840$ is the same as Ensemble 1.
penev_B_F00497	14	■ 0.992	$10\times$	$10\times$	The image was converted to grayscale, oversampled $10\times$, and then under-